Certified Adversarial Risk Bounds for Quantum Error-Correcting Memories: Injection-Budget Certification with Simultaneous Intervals and Reproducible Three-Code Bivariate-Bicycle Benchmarks

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QEC Emulator v1.2.4 [1] (ldpc 2.4.1, pymatching 2.2.2, stim 1.16.0, numpy 2.5.2)

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Abstract

We certify *adversarial risk bounds* for quantum error-correcting memories: for a fixed implementation (code + schedule + decoder) and a bounded-weight fault-injection adversary, the operational risk $\lambda_{\mathcal{M}}(W) = \max_{|F|\le W} \Pr[\text{logical failure} \mid F]$ is estimated on device and reported as simultaneous intervals $[L_W, U_W]$ at joint $\geq 95\%$ confidence (Bonferroni-simultaneous; exact Clopper–Pearson at zero counts), with honest $U_W = 1$ wherever the weight region is not exhausted. The theory pairs two acknowledged background tools—a channel-game minimax saddle point and a Knill–Laflamme fidelity floor—with new elements: a spacetime-budget correctability theorem with sharpness and device-aware composition, extended to CSS memories, plus a corrected Pauli-channel bridge from either-sector logical failure rates to diamond-norm bounds. Experiments use the open-source QEC Emulator v1.2.4 with deterministic seeds and SHA-256 provenance. Across three bivariate-bicycle codes: an exhaustive BB[[72,12,6]] certificate ($d = 6$ exactly; published witnesses, independent checker); benchmarks in which BP+OSD lies far below two graphified matching baselines (risk ratios 67.8× to >134,000×, joint $\geq 95\%$); a pinned seven-decoder paired grid on 70,000 bit-identical instances (BP augmentation closes the gap to BP+OSD; bare symmetry matching is non-viable); circuit-level corroboration on one pinned operating point, where Relay-BP separates from BP+OSD (the two are statistically tied at code capacity); and a five-decoder re-analysis of the released Kunlun [[18, 4, 4]] detector model. On ibm_marrakesh, calibration-informed placement lowered logical error in all four configurations ($p = 0.042$); an injection test with protocols, masks, and decision rules hash-pinned before submission confirmed the distance cliff (92–96% collapse) and the device-aware composed bound on every tested arm, replicating across calibration cycles, devices, and distance 7; stochastic cells meet mask-conditioned thresholds in both campaigns (the conditioning estimand was corrected retrospectively in campaign 1 and pinned before submission for the consolidation; chronology disclosed in Appendix C). These results assemble into a scoped integrity-evidence profile with an explicit threat model; syndrome-only monitoring is proved insufficient.

Keywords: qLDPC; bivariate-bicycle codes; adversarial channels; minimax; entanglement fi-

delity; BP+OSD; MWPM; DEM decomposition; reproducible benchmarks; open-source scientific software.

1 Introduction

Fault-tolerant quantum computing and post-quantum cryptography are strategically coupled [2–5]. The bivariate-bicycle (BB) code family [6] provides a high-rate, weight-six qLDPC substrate suitable for both defensive deployment and adversarial analysis.

Specific gap. Prior BB benchmarks [6, 7] use BP+OSD; this paper adds a transparent graphified DEM-MWPM reference baseline compatible with PyMatching-style graphlike inputs, together with a three-code same-protocol reproducibility archive with SHA-256 manifests. We fill these gaps for the tested implementations and find that the two graphified baselines perform substantially worse than BP+OSD under this protocol.

1.1 What this paper claims (precise)

(i) **Certified adversarial risk bounds as the central object.** The fixed-implementation adversarial risk $\lambda_{\mathcal{M}}(W)$ (Definition 4), reported as simultaneous intervals $[L_W, U_W]$ with honest $U_W = 1$ outside tested regions (Remarks 1, 2); a four-category decoder endpoint separating wrong-logical corrections from invalid ones (Definition 3).

(ii) Background (acknowledged standard, not a contribution): minimax saddle point via Sion's theorem; Knill–Laflamme fidelity floor.

(iii) Architectural scaling lemma on logical-operator weight.

(iv) QEC Emulator v1.2.4 — open-source software infrastructure for all experiments, with five independently extensible layers and SHA-256 provenance.

(v) BB72 certificate: $d = 6$ exact, $t = 2$ exhaustive, SHA-256 matrix pins.

(vi) Three-code single-round BP+OSD vs DEM-MWPM, 10,000 trials, CIs, with a 10^6-trial rare-event refinement at $p \in \{0.002, 0.004\}$.

(vii) Three-code repeated-cycle BP+OSD (d rounds), 2,000–500 trials, CIs, plus 4,000–20,000-trial adaptive extensions.

(viii) Three-code finite-size evidence (sub-threshold ordering, not formal FSS).

(ix) Hardware ablation (simulation models only): SWAP routing, measurement noise, and gate-decoherence parameters materially affect the simulated CZ/MS gap; no real-hardware causal claim is made.

(x) UF-BFS provides a second graphified baseline; it also underperforms BP+OSD under the submitted protocol.

(xi) Decoder-dependent ACE gap (Remark 11; upgraded from proxy to bound under the Pauli approximation by Proposition 1).

(xii) Injection-budget game: operational ACE value (Definition 4), spacetime-budget correctability theorem with sharpness construction (Theorem 3), device-aware composition bound (Corollary 3).

(xiii) Device test of the theory layer on ibm_marrakesh with protocols, masks, and decision rules hash-pinned before submission (Appendix B, 312 arms, 180,000 shots): weight-t worst-case injections bounded, weight-$(t+1)$ collapse at 92–96%; all stochastic cells hold against mask-conditioned thresholds in both campaigns (campaign-1 conditioning estimand corrected retrospectively — Appendix C); the device-aware composed bound holds on all tested arms.

(xiv) No asymptotic threshold; no IND-CPA.

2 Related work

BB codes and their threshold [6]; networked BB codes [7]; BP+OSD [8, 9]; PyMatching/Sparse Blossom [10, 11]; Stim [12]; qLDPC families [13–15]; FTQC overhead [16–18]; post-quantum standards [3–5]; QKD [19–21]; formal verification [22–25].

Decoder landscape, 2025–2026. Relay-BP, a BP variant with disordered memory strengths targeting real-time circuit-level decoding [26] with an FPGA prototype on $[[144, 12, 12]]$ [27]; symmetry matching (`symatch`), which exploits the equivalence of BB codes to toric-code copies [28]; belief-matching hybrids that reweight matching graphs with BP posteriors [29]; trapping-set analyses of detector error models [30]; and the below-threshold Λ-ladder reporting standard of Acharya et al. [31]. On hardware, the Kunlun processor operated a $[[18, 4, 4]]$ BB code and released its detector matrices openly [32], and a controlled study of defective components contrasted defect exclusion with defect-aware decoding [33].

Closest prior art. Table 1 positions this work against the nine closest sources. The strongest comparison is the Fail-fast framework of [34]: both works inject structure to probe rare QEC failures, but the estimands differ — fail-fast accelerates *nominal stochastic* rare-event estimation (importance-style variance reduction under the device's own noise law), whereas we bound *worst-case adversarial* risk $\lambda_{\mathcal{M}}(W)$ over fault-set placement with simultaneous $[L_W, U_W]$ intervals; the two are complementary, and fail-fast sampling is a candidate accelerator for our U_W estimation at large W. The trapping-set analysis of [30] is closest in spirit on decoder failure structure; it characterizes failure configurations of a *stochastic* DEM, while our injection-boundary study fixes the code and varies an *adversarial* weight budget, separating wrong-logical corrections from residual-carrying invalid corrections (four-category endpoint) — a distinction the trapping-set frame does not track.

Table 1: Closest prior art and differentiation. Scope abbreviations: nom. = nominal stochastic noise; adv. = adversarial placement.

source	contribution	relation to this work
Bravyi et al. [6]	BB codes, thresholds, hardware memory	we reuse the code family; they do not treat adversarial injection
Roffe et al. [8]	BP+OSD decoder	our reference decoder; benchmarked, not introduced here
Sahay et al. [28]	symmetry matching for BB codes	pinned-comparator subject; bare form non-viable on this instance, BP-augmented tied with BP+OSD
Müller et al. [26]	Relay-BP real-time BP	benchmarked; circuit-level outcome separates from the code-capacity tie
Beverland et al. [34]	fail-fast rare-event probing (nom.)	strongest comparison; accelerated nom. estimation vs our adv. worst-case bounds; complementary
Pacenti et al. [30]	DEM trapping sets	stochastic failure structure; no adversarial budget or endpoint decomposition
Kosut et al. [35]	recovery optimization for a fixed channel	defender-side optimization; our $\lambda_{\mathcal{M}}(W)$ fixes the implementation
Huang et al. [36]	coherent-error channel analysis	basis of our twirling residual, which we close analytically
Acharya et al. [31]	below-threshold Λ reporting	reporting standard we delimit our block-Λ against (no matched protocol)

Section 6 positions the present work against these families quantitatively.

3 QEC Emulator: software infrastructure

All experiments in this paper are conducted using the QEC Emulator v1.2.4 [1], an open-source Python package released under the MIT licence at doi:10.5281/zenodo.20588043 (published June 2026). This section describes the emulator's design, scientific contributions, and its role as a reusable research platform.

3.1 Design philosophy and architecture

The emulator is organised as five independent layers (Figure 1), each with a single scientific responsibility:

L1. Code construction (`codes.py`) — BB code algebra over $\mathbb{F}_2$, SHA-256 matrix fingerprints.

L2. Decoders (`decoders.py`) — LookupDecoder, BPOSDDecoder, MWPMDecoder; common abstract interface.

L3. Noise models (`noise.py`) — four models from phenomenological to hardware noise-model simulation.

L4. Benchmark runners (`benchmark.py`) — five runners covering all experimental protocols in this paper.

L5. Provenance (`provenance.py`) — SHA-256 integrity seals, manifest export, QC checks.

The layers are strictly decoupled: adding a new decoder, noise model, or benchmark runner requires subclassing or implementing a single interface without modifying any other layer.

L1. Code construction
`BBCode(ell, m, a_exp, b_exp)` $\rightarrow H_X, H_Z$ + SHA-256 pins
L2. Decoders
`LookupDecoder` | `BPOSDDecoder` | `MWPMDecoder`
`BaseDecoder`: `decode_z(syn)` $\rightarrow$ correction mask
L3. Noise models
`PhenomenologicalModel` | `RepeatedCycleModel`
`CircuitLevelModel` | `HardwareModel(CZ|MS)`
L4. Benchmark runners
`run_sweep` | `run_repeated_cycle` | `run_threshold_scan`
`run_fixed_weight` | `run_distance_certificate`
L5. Provenance
`ProvenanceRecord`: SHA-256(code, decoder, p, seed, failures)
`ProvenanceLogger`: CSV/JSON export, `unresolved_pairs()` QC check

Figure 1: Five-layer architecture of QEC Emulator v1.2.4. Each layer has a single scientific responsibility and is independently extensible.

3.2 Code construction layer

The `BBCode` dataclass constructs BB codes from generator polynomial exponents over $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$. Three named constructors are pinned to this paper's three code instances:

Constructor	(ℓ, m)	Polynomials	n	k	H_X SHA-256 prefix
`BB72()`	(6, 6)	$A = x^3 + y + y^2$, $B = y^3 + x + x^2$	72	12	`7ab0973b...`
`BB144()`	(12, 6)	same polynomials	144	12	`195d5586...`
`BB288()`	(12, 12)	$A = x^3 + y^2 + y^7$, $B = y^3 + x + x^2$	288	12	`c06e721e...`

The BB288 polynomial parameters follow Bravyi et al. Table 1 [6]. Every `BBCode` instance exposes `hx_sha256()` and `hz_sha256()` methods returning the full 64-character SHA-256 digest of the respective matrix's byte representation. The full digests are:

Code	H_X SHA-256
BB72	`7ab0973bfd02e399d69728d26e67dbfa0e95bbc6ad2c5cf8bf309d667244a5a8`
BB144	`195d5586a406f0194b89649b6a2499ec2b481edb81c91293daf1745d8b806641`
BB288	`c06e721ec6ea114b60e0e6f7187324f6e150e229b8d008888e8a4cc3f4a324a2`

A reviewer can verify any data table in this paper by (i) running `pip install qec-emulator`, (ii) instantiating the named constructor, and (iii) comparing `hx_sha256()` against the digest above. This creates a tamper-evident chain from published number back to exact matrix bytes.

3.3 Decoder layer and key discoveries

Three decoder classes share a common abstract base interface (`decode_z`, `decode_x`, `decode_success`), making them interchangeable in all benchmark runners.

`LookupDecoder`. Exact radius-t syndrome lookup, self-contained with no external dependencies. For BB72 with certified $d = 6$ and $t = 2$, the lookup table covers all $\sum_{w=0}^{2} \binom{72}{w} = 2{,}629$ weight-≤ 2 patterns; the `run_fixed_weight` runner confirms zero failures over all patterns exhaustively.

`BPOSDDecoder`. Wraps the `ldpc` package v2 `BpOsdDecoder` class API [9]. The legacy `bposd_decoder` function (v1 API) causes a segmentation fault on some inputs in `ldpc` ≥ 2.0 and must not be used. This is a non-obvious breaking change; the emulator explicitly uses the v2 class API and documents the reason. Default parameters: min-sum BP, scaling factor 0.625, OSD-CS order 7, 100 maximum iterations. If `ldpc` is absent the class falls back to `LookupDecoder` with `is_fallback=True`, so provenance records are never silently mislabelled.

`MWPMDecoder` and the DEM chain decomposition. This is an important implementation detail documented here. PyMatching v2 [10] strictly rejects parity-check matrix columns with weight > 2 (raises `ValueError`). BB codes have column weight three: each data qubit triggers exactly three syndrome checks. Therefore, loading H_X directly into PyMatching v2 crashes.

The emulator implements the DEM chain decomposition: each weight-3 column $\{a, b, c\}$ is split into two binary edges $\{a, b\}$ and $\{b, c\}$ via the `add_edge()` API, discarding the $\{a, c\}$ correlation. Additionally, BB syndrome graphs can contain odd-parity connected components arising from logical operators, which prevent perfect matching; high-weight (100× log-likelihood) global boundary edges on every detector node resolve this.

The experimental consequence (Section 5) is large but deliberately limited: the tested DEM-chain MWPM reference baseline is 67.8× to >1,000× worse than BP+OSD across the tested BB code instances (BB72 finite; BB144/BB288 infinite where BP+OSD achieves zero failures). This should not be read as a theorem about all matching-family decoders. It shows that the submitted chain graphification is a lossy, reproducible reference baseline; any future MWPM-family comparison for BB codes must state precisely how weight-3 hyperedges, correlations, and boundary effects are represented.

3.4 Noise model layer

`PhenomenologicalModel`. Independent Bernoulli Pauli errors at rate p, ideal syndrome readout. Used for all single-round decoder comparisons.

`RepeatedCycleModel`. The v1.2.4 protocol required for proper multi-round benchmarking. d syndrome rounds per trial: rounds $1, \ldots, d-1$ apply independent Bernoulli errors at rate p to both data qubits and measurement bits; round d is noiseless. Returns differential detector syndrome arrays of length $d \times n_{\text{checks}}$ for space-time decoders. This model is not equivalent to d independent single-round trials: cumulative data errors and measurement noise correlate across rounds in a way that single-round simulation cannot capture.

`CircuitLevelModel`. Single-cycle Pauli propagation through the BB 432-CNOT schedule with ancilla preparation faults (p_{prep}), gate faults (p_{gate}), idle data-qubit decoherence ($p_{\text{idle}} = 0.05\, p_{\text{gate}}$ by default), and noisy syndrome measurement (p_{meas}).

`HardwareModel`. Architecture-parameterised single-cycle simulation for two backends: CZ (superconducting, $T_1 = 150\,\mu$s, SWAP routing) and MS (trapped-ion, $T_1 > 1$ s, all-to-all connectivity). Device parameters match published characterisations.

3.5 Benchmark runner layer

Five runners implement the complete experimental protocols of this paper.

`run_sweep`: single-round sweep across a p-grid; one `ProvenanceRecord` per (p, batch) pair.

`run_repeated_cycle`: proper repeated-cycle benchmark. For `decoder_type='bposd'`, constructs the space-time extended check matrix:

$$H_{\text{ext}} = [H_{\text{space}} \mid H_{\text{meas}}], \quad H_{\text{space}} = I_d \otimes H_X, \tag{1}$$

where H_{meas} captures consecutive-round measurement-flip detection. For `decoder_type='mwpm'`, builds a space-time matching graph with per-round data-qubit edges and between-round measurement-flip edges.

`run_threshold_scan`: multi-code finite-size comparison.

`run_fixed_weight`: fixed-weight random-sampling decoder policy test (not a code-capability test beyond the certified radius).

`run_distance_certificate`: meet-in-the-middle exhaustive distance lower-bound enumeration.

3.6 Provenance layer

The provenance layer is the core scientific contribution of the emulator. Every `ProvenanceRecord` stores 15 fields including the SHA-256 of H_X, the SHA-256 of H_Z, the integer seed, the trial count, the raw failure count, the Wilson 95% upper confidence bound, the UTC timestamp, the Python/NumPy version, and an integrity seal:

$$\texttt{result_sha256} = \text{SHA256}(\text{JSON_sort}(\text{code, decoder, } p, n_{\text{trials}}, \text{seed, failures, } \texttt{hx_sha256}, \texttt{hz_sha256})). \tag{2}$$

Any change to the code matrix, seed, trial count, or failure count produces a completely different digest, making undetected manipulation cryptographically infeasible. The `unresolved_pairs()` method flags any two records sharing an identical `(code, decoder, p, result_sha256)` tuple — distinguishing genuine zero-failure runs from seed collisions or copy-paste errors.

3.7 Open-source design and research extensibility

QEC Emulator is released under the MIT licence and is explicitly designed as a platform for further research. Four extension points require no modification to the core package:

1. **New BB code instance**: `BBCode(ell, m, a_exp, b_exp)`.
2. **New CSS code**: override `hx` and `hz` in a `BBCode` subclass.
3. **New decoder**: subclass `BaseDecoder` and implement `decode_z` and `decode_x`.
4. **New noise model**: implement the four-return-value callable `(code, p, rng)` $\rightarrow$ `(z_mask, x_mask, syn_z, syn_x)`.

Directions for academic extension include: threshold analysis with ≥ 3 code sizes using `run_repeated_cycle`; hyperedge-aware MWPM once an open-source hypergraph matching solver becomes available; new hardware backends in `HardwareModel.PRESETS`; networked BB code simulation following [7]; and use of the provenance layer as an evidence record in formal quantum cryptographic infrastructure audits.

3.8 Testing and continuous integration

The package ships with 52 pytest tests (50 active, 2 correctly skipped when optional dependencies are absent). Tests cover all five layers: code construction, decoder correctness, noise model output shapes and limits, provenance integrity, and benchmark runner determinism. Critical regression tests include:

- No segmentation fault from the ldpc v2 API (`BpOsdDecoder`).

- No `ValueError` from PyMatching v2 on weight-3 BB columns.
- Odd-parity syndrome decoded without exception (50 random trials).
- `result_sha256` integrity seal changes when any input changes.

Tests run on Python 3.9, 3.11, and 3.12 via GitHub Actions (`.github/workflows/ci.yml`).

3.9 Minimal usage example

Listing 1: Single-round sweep with provenance logging (QEC Emulator v1.2.4).

```
from qec_emulator import BB72, BPOSDDecoder, PhenomenologicalModel
from qec_emulator.benchmark import run_sweep

logger = run_sweep(
    BB72(), BPOSDDecoder(BB72()), PhenomenologicalModel(),
    p_values=[0.004, 0.006, 0.008, 0.010, 0.012],
    trials=10000, seed=20260607,
)
logger.to_csv("bb72_results.csv")
assert len(logger.unresolved_pairs()) == 0 # QC: no seed collisions
```

Listing 2: Repeated-cycle benchmark across three code sizes (v1.2.4 protocol).

```
from qec_emulator import BB72, BB144, BB288
from qec_emulator.benchmark import run_repeated_cycle

for code, rounds in [(BB72(), 6), (BB144(), 12), (BB288(), 18)]:
    logger = run_repeated_cycle(
        code, decoder_type='bposd',
        p_values=[0.004, 0.006, 0.008, 0.010, 0.012],
        rounds=rounds, trials=2000, seed=20260607,
    )
    logger.to_json(f"rc_bposd_{code.n}.json")
```

4 ACE framework

The defensive strategy is $\mathcal{D}_\phi = \mathcal{R}_\phi \circ \mathcal{S}_\phi \circ \mathcal{T}_\phi \circ \mathcal{Q}_\phi$. Adversarial class $\mathfrak{A}_1$: Pauli-weight-bounded (theorems); $\mathfrak{A}_2$: circuit-level stochastic (experiments); $\mathfrak{A}_3$: cryptanalytic (outside all theorems).

Theorem 1 (Minimax existence)**.** *Let $\mathfrak{D}$, $\mathfrak{A}$ be nonempty compact convex CPTP sets on finite-dimensional $\mathcal{H}$. Then* $\max_{\mathcal{D}} \min_{\mathcal{A}} F_e = \min_{\mathcal{A}} \max_{\mathcal{D}} F_e$ *(Sion [37]; CPTP sets compact convex in Choi-matrix space).*

Role of Theorem 1. The saddle-point result is used only to guarantee that the operational value of Definition 4 is well-defined; every quantitative statement below is a bound on that value, not a property of the saddle itself.

Theorem 2 (Fidelity floor)**.** *For $\mathcal{C} = [[n, k, d]]$ with $t = \lfloor (d-1)/2 \rfloor$, ideal recovery $\mathcal{R}_\mathcal{C}$ gives $F_e = 1$ for all $\mathcal{A} \in \mathfrak{A}_1$. Implemented recovery within diamond-norm error $\varepsilon_{\mathrm{FT}}$ gives $F_e \geq 1 - \varepsilon_{\mathrm{FT}}$ and $V^\star \geq 1 - \varepsilon_{\mathrm{FT}}$ [38, 39].*

Corollary 1. *For $d_n \geq \delta_0 n$, every attack of weight $w_n \leq \lfloor (d_n - 1)/2 \rfloor \approx \delta_0 n/2$ is correctable.*

4.1 The injection-budget game: an operational ACE value

Definition 1 (Spacetime fault; weight)**.** *For a memory experiment with n_d data qubits and R syndrome rounds, a* spacetime fault set *F is a set of elementary faults: (i) one Pauli operator applied to one data qubit at one inter-round location; (ii) one faulty syndrome measurement outcome. The weight $|F|$ counts elementary faults. F is* adversarially placed *when its support is chosen arbitrarily in space and time.*

Definition 2 (Injection-budget adversary $\mathfrak{A}_1^{\mathrm{st}}(W)$)**.** *The adversary selects any injected fault set F with $|F| \le W$, with full knowledge of code, schedule, and decoder, no access to syndrome outcomes before selection (non-adaptive), and no influence on native noise $N \sim \mathcal{P}_{\mathrm{dev}}$ or on decoding.*

Definition 3 (Four-category decoder endpoint)**.** *For a single shot with true residual error r_{true} (injected $\oplus$ native) and observed syndrome s, a decoder answer $\hat{c}$ is classified as:* valid_correct *if $H\hat{c} = s$ and $\hat{c} \oplus r_{\mathrm{true}}$ lies in the stabilizer group;* valid_wrong_logical *if $H\hat{c} = s$ and $\hat{c} \oplus r_{\mathrm{true}}$ is a nontrivial logical operator (a decoder logical failure);* invalid_residual *if $H\hat{c} \neq s$ (no syndrome-consistent correction was produced — a decoder failure, not a code failure); and* abort *if the decoder returns no answer (timeout, infeasible matching, or exception). Throughout this manuscript, headline rates decompose into these categories; the security-relevant endpoint is* valid_wrong_logical*, and* ***`invalid_residual`*** *and* ***`abort`*** *are reported separately, never counted as logical failures (this repairs the v23 endpoint conflation, under which e.g. all* 2,332 *UF-BFS weight-two failures carried nonzero residual syndrome).*

Definition 4 (Operational ACE value)**.** *For implemented memory $\mathcal{M}$ (code + schedule + decoder),*

$$\lambda_{\mathcal{M}}(W) := \max_{F:\,|F|\le W} \Pr_{N}\big[\textit{logical failure} \mid F, N\big], \qquad V_{\mathcal{M}}(W) := 1 - \lambda_{\mathcal{M}}(W).$$

$\lambda_{\mathcal{M}}(W)$ is directly measurable: inject F on the device, decode, count logical failures (Appendix B).

Remark 1 (Central estimand of this work)**.** *Definition 4 is the central object of this paper: a* fixed-implementation *adversarial risk, the worst case $\lambda_{\mathcal{M}}(W) = \max_{F \in \mathfrak{A}_1^{\mathrm{st}}(W)} q(F)$ of the conditional logical-failure probability $q(F)$ over the adversary's fault-set choice, for one concrete implementation $\mathcal{M}$ (code + schedule + decoder + device noise model). Quantitative results below are explicitly labelled as one of three distinct objects, which are not interchangeable:* (i) nominal averages *over a noise model (the decoder grids and repeated-cycle ladders) — these are* not *estimates of $\lambda_{\mathcal{M}}(W)$ and are never reported as such;* (ii) per-attack risks *$q(F)$ for individually tested fault sets, each with its own interval; and* (iii) worst-case intervals *$[L_W, U_W]$ for $\lambda_{\mathcal{M}}(W)$ itself, constructed only where an attack set is pinned (Remark 2, Table 2). By contrast, the saddle-point statement of Theorem 1 and the Knill–Laflamme-style fidelity floor of Theorem 2 are background: they certify that channel-game quantities are well-defined and that ideal recovery is perfect inside the correctable radius, but no saddle-point property is invoked in any bound below, and no theorem optimizes over defensive strategies. Optimizing $\mathcal{D}$ (code and decoder co-design against $\mathfrak{A}_1^{\mathrm{st}}$) is an open problem we do not address.*

Remark 2 (Reporting convention: worst-case intervals $[L_W, U_W]$)**.** *Wherever this work reports an interval for the worst-case risk $\lambda_{\mathcal{M}}(W) = \max_{|F|\le W} q(F)$ over a pinned attack class, the endpoints obey the following rules. L_W is the largest lower confidence bound attained by a* tested *fault set of weight $\le W$ (adversarial evidence: some tested F attains at least this risk; the maximum over tested sets of valid lower bounds is a valid lower bound on the maximum). U_W is valid only in one of two cases:* (i) exhaustive testing *— if the tested sets exhaust the weight-$\le W$ region and the per-set upper bounds hold simultaneously, then $U_W = \max_{F\,\mathrm{tested}} U(F)$, the*

maximum *of the per-set upper bounds. Taking the minimum of arm-specific upper endpoints is invalid: attacks with true risks* 0.1 *and* 0.9 *and valid individual upper endpoints* 0.2 *and* 0.95 *would give* min $= 0.2 < 0.9$*, failing to bound the worst attack.* (ii) incomplete testing — *if the budget region contains untested fault sets,* $U_W = 1$ *is reported honestly, unless a proved analytic bound covers the untested region; silence is not evidence of correctness. Where a certificate exhausts the weight region (O4; the noiseless exhaustive radius test of Table 3),* $L_W = U_W$ *is the certified value exactly. Simultaneity is claimed* per family*: each displayed table defines its family of cells, and cell intervals are Bonferroni-adjusted within that family only; no manuscript-wide familywise guarantee is made. Exact Clopper–Pearson bounds are used at zero counts (Remark 12). The adaptive repeated-cycle ladder (Section 7) uses a pinned stopping rule; its intervals are time-uniform confidence sequences [40] computed by the archived statistics module (`stats_intervals.py`), valid under the sequential stopping — fixed-sample formulas would not be valid there (a nominal 95% Wilson interval under that stopping rule covers* $\approx 93.9\%$ *at* $p = 0.021$*). All interval-producing numbers regenerate from `stats_intervals.py` with an explicit estimand, sampling unit, stopping rule, family definition, and confidence allocation per table.*

Table 2: Risk-certificate inventory (Remark 2). One row per (system, budget) for which a worst-case interval $[L_W, U_W]$ for $\lambda_{\mathcal{M}}(W) = \max_{|F|\le W} q(F)$ is reported anywhere in this work. L_W is the maximum over *tested* fault sets of valid lower bounds; U_W is the maximum of simultaneous per-set upper bounds where testing is exhaustive (exact rows, $L_W = U_W$) and 1 wherever untested fault sets remain (rule (ii); silence is not evidence of correctness). Interval cells use exact Clopper–Pearson bounds with one Bonferroni family over all $K = 676$ interval cells of this table ($\alpha = 0.05$); exhaustive and witness rows carry no statistical error. The noiseless weight-3 row is certified by 830 exhibited deterministic witnesses (the sharpness direction of Theorem 3(b), realised on BB72). "tested" counts distinct masks across both memory bases. Files: `data/risk_certificate_v24.csv`, regenerated by `gen_risk_certificate_v24.py` from `inj_radius_noiseless_v24.csv`, `injection_boundary_decoders_v24.csv`, and the two device campaign result tables (Appendix B).

system	noise model	W	$\lvert\mathfrak{A}(W)\rvert$	tested	allocation	L_W	U_W	gap
BB72+BP+OSD	noiseless	2	5,258	exhaustive	exact enum.	0	0	0
BB72+MW-lookup	noiseless	2	5,258	exhaustive	exact enum.	0	0	0
BB72+UF	noiseless	2	5,258	exhaustive	exact enum.	1	1	0
BB72+BP-guided	noiseless	2	5,258	exhaustive	exact enum.	1	1	0
BB72+BP+OSD	noiseless	3	119,280	40,016 (33.6%/sector)	exact (witnesses)	1	1	0
BB72+BP+OSD	phenom. p=0.003	2	5,258	12 (0.23%)	CP, Bonf.-676	0.0147	1	0.9853
BB72+BP+OSD	phenom. p=0.003	3	119,280	8 (0.007%)	CP, Bonf.-676	0.8857	1	0.1143
REP5 (ibm_marrakesh)	device	2	32	15 (46.9%)	CP, Bonf.-676	0.0587	1	0.9413
REP5 (ibm_marrakesh)	device	3	52	19 (36.5%)	CP, Bonf.-676	0.9703	1	0.0297
REP7 (ibm_marrakesh)	device	4	198	5 (2.5%)	CP, Bonf.-676	0.9617	1	0.0383
REP3 (ibm_marrakesh)	device	2	14	5 (35.7%)	CP, Bonf.-676	0.9185	1	0.0815

Theorem 3 (Spacetime-budget correctability and sharpness — protected sector)**.** *Let* $\mathcal{M}$ *be the distance-d (d odd) repetition memory with* $R \ge 1$ *rounds, decoded by* minimum-cardinality *matching on the spacetime detector graph (equivalently, minimum-weight matching with uniform edge weights);* $t = \lfloor (d-1)/2 \rfloor$*.* ***Scope:*** *the fault set is restricted to the* protected sector — *bit-flip (X) data faults and measurement faults for the Z-basis memory. The bare repetition code has no phase protection: a single Z fault on one data qubit is a logical phase error, so the all-Pauli version of part (a) is false and is not claimed (this restricts the v23 statement, which was read against the unrestricted fault set of Definition 1). Per-theorem scope in this manuscript: classical observable / one CSS sector / full channel are stated separately for each result.*

(a) Ideal extraction. *If* $N = \varnothing$ *then* $\lambda_{\mathcal{M}}(W) = 0$ *for all* $W \le t$*.*

(b) Sharpness. *An explicitly constructible injection with $|F| = t+1$ makes minimum-cardinality matching fail with probability one: $\lambda_{\mathcal{M}}(t+1) = 1$ already at $N = \varnothing$.*

(c) Device-aware composition. *For arbitrary native noise, writing $\pi(k) := \Pr[|N| \geq k]$ ($\pi(k) = 1$ for $k \leq 0$),*

$$\lambda_{\mathcal{M}}(W) \;\leq\; \pi\big((t+1) - W\big) \qquad (0 \leq W \leq t).$$

Proof. In the spacetime detector graph [41, 42], elementary faults are edges: data faults are spatial edges and measurement faults are temporal edges. Any logical chain connects the two spatial boundaries and contains $\geq d$ spatial edges; temporal edges do not advance the spatial coordinate. (a) A fault set F flips the boundary of its chain; the decoder returns a minimum-cardinality chain C with the same boundary, $|C| \leq |F|$. Failure requires $F \oplus C$ to contain a logical chain, so $|F| + |C| \geq d$, hence $|F| \geq t{+}1$. (b) Inject X on $t{+}1$ consecutive central data qubits in one round: the boundary completion has weight $d - (t{+}1) = t < t{+}1$, is chosen, and $F \oplus C$ is the full logical string. (c) Faults compose by symmetric difference (Pauli/binary composition), not set union: $E = N \oplus F$, with $|E| \leq |N| + |F|$. Failure $\Rightarrow |E| \geq t{+}1 \Rightarrow |N| \geq t{+}1 - W$; take probabilities. (Cancellation of an injected and a native fault at the same location only shrinks $|E|$, so the bound is conservative.) *Weighted decoders.* The cardinality hypothesis is what the proof uses: for a decoder minimizing a general edge-weight functional $w(\cdot)$ (e.g. negative log priors), the identical argument with $w(C) \leq w(F)$ gives failure only when $2\,w(F) \geq \min_{L\ \text{logical}} w(L)$, i.e. a correctability radius in the decoder's own fault metric rather than in cardinality; we do not claim the cardinality radius for weighted decoders, and all prior-weighted MWPM results in this manuscript are numerical, not deduced from this theorem. □

Remark 3 (Scope and BB72 support)**.** *The proof uses only the chain complex and minimum-cardinality decoding, so (a) extends to any CSS memory whose spacetime distance is d under the same decoder class (uniform weights; see the weighted-decoder caveat in the proof). For BB72, the single-round analogue at $W \leq t = 2$ is established* exhaustively *by the finite-instance certificate (Section 10): every weight-≤ 2 Pauli pattern is corrected. A general spacetime-distance proof for BB memories is open.*

Remark 4 (Scope of injection certification — three honest outputs)**.** *The v23 version of this statement claimed a* sound and complete *certification test; that claim is withdrawn. Under the spacetime fault model with exact minimum-weight decoding, the pinned injection protocol of Appendix B yields three distinct, separately scoped outputs.* (i) Worst-case certificate on an enumerated attack set. *Over $W_{\leq t}$ (all $\binom{n}{\leq t}$ single-sector patterns) the finite-instance certificate gives $\lambda_{\mathcal{M}}(W) = 0$ exactly, with the coverage argument stated explicitly (Section 10). Soundness direction: if any weight-$\leq t$ injection arm exhibits a logical failure under ideal extraction, the implementation is faulty, since Theorem 3(a) gives $\lambda_{\mathcal{M}}(W) = 0$ for the model-correct memory.* (ii) Model-conditional bound. *Under the composed Binomial⊗Poisson native-fault model, the stochastic bound of Corollary 3 holds conditionally on that model and its calibration interval; it is a model prediction, not a device property (Section 12).* (iii) Challenge test restricted to the tested set. *If the memory is vulnerable at weight $t{+}1$, the explicitly constructible patterns of Theorem 3(b) exhibit the vulnerability with probability one under ideal extraction — a falsifiable challenge for the patterns tested. Coverage of untested masks, mixed Paulis, arbitrary spacetime placement, and measurement faults is an explicit gap; on any unexplored region the honest upper bound is $U_W = 1$.*

Corollary 2 (CSS spacetime correctability)**.** *Let $\mathcal{C}$ be a CSS code with X- and Z-distances d_X, d_Z, and let $\mathcal{M}$ be its R-round memory experiment under the spacetime fault model of Definition 1, decoded by exact minimum-weight spacetime decoding. Then any adversarially placed fault set F with $|F| \leq t$, where $t = \lfloor(\min(d_X, d_Z) - 1)/2\rfloor$, causes zero logical failures under ideal extraction.*

Proof. Apply the chain argument of Theorem 3(a) to each CSS component independently; X-type faults couple only to Z-type checks and vice versa. The decoder returns a minimum-weight correction C with the same syndrome as F, so the residual $F \oplus C$ has weight $\leq |F|+|C| \leq 2|F| \leq 2t < \min(d_X, d_Z)$. A nontrivial logical operator of either sector has weight $\geq \min(d_X, d_Z)$, and measurement faults contribute only temporal edges, which do not change the data-qubit residual's logical class; hence $F \oplus C$ is not a nontrivial logical and no logical failure occurs. □

Remark 5 (Decoder hypothesis)**.** *For codes whose checks have weight > 2 the spacetime detector complex is a hypergraph, so Corollary 2 concerns exact minimum-weight decoding, not matching on a graphified reduction — the distinction measured in Sections 5 and 6. The fault model is the spacetime model of Definition 1; circuit-level hook errors, in which one ancilla fault propagates to several data faults, can lower the effective spacetime distance and are outside its scope, which is why the BB72 circuit-level evidence rests on the exhaustive certificate and the circuit-level simulations rather than on this corollary.*

Remark 6 (Decoder instantiation of Corollary 2)**.** *Exact minimum-weight decoding is computationally intractable at BB72 scale, so it is worth stating precisely which conclusions survive the replacement of the reference decoder by an implemented one.* Within the certified radius*, correctability is a property of the code, not of the decoder: any decoder that corrects every weight-$\leq t$ pattern achieves the same zero-failure region, and for BB72 the certificate (O4) verifies this exhaustively for the implemented radius-2* `LookupDecoder` *(*2,629/2,629 *patterns, single round). Corollary 2's conclusion is therefore inherited by a concretely implemented decoder on the certified region.* Beyond the radius *the corollary is silent by construction, and the manuscript's beyond-radius BB72 claims (Sections 5 and 7) are empirical: Theorem 3(c) and Corollary 3 bound the reference exact decoder, while the measured BP+OSD failure rates with Wilson intervals quantify the realised implementation. No π-bound is asserted for BP+OSD, and no optimality ordering between minimum-weight and maximum-likelihood coset decoding is assumed — on degenerate codes the two can differ, which is why the certificate's checks are decoder-explicit.*

Remark 7 (Cross-decoder calibration of the injection boundary)**.** *Remark 6 separates certified from empirical conclusions; we quantify the empirical side across three decoder families plus a minimum-weight lookup reference by replaying the injection probe matrix — controls of weight $W \leq 2$ versus the O5-certified weight-3 halves of weight-6 logicals (Corollary 4) — in the single-round phenomenological model at $p = 0.003$ (8,000 trials per control, 4,000 per certified half). BP+OSD preserves the certified boundary: maximum control wrong-logical rate* 1.96% *(X) /* 0.16% *(Z) versus certified pair-maxima* 73–87% *(*`run_inj_boundary_decoders_v24.py`*;* `data/injection_boundary_decoders_v24.csv`*,* `data/inj_radius_noiseless_v24.csv`*). The v24 rebuild of this study (review Sec. 5.3) freezes the probe set once, replays* identical *native-error realisations to every decoder, separates a noiseless exhaustive radius test from the noisy injection experiment, and decomposes every outcome into the four-category endpoint of Definition 3. Two findings survive the stricter protocol and one v23 claim is corrected:* (i) *on the noiseless exhaustive radius test, BP+OSD and the minimum-weight lookup reference commit zero wrong-logical corrections on all* 2,629 *weight-≤ 2 masks per sector, while Union-Find commits* 13 *(X) and BP-guided MWPM* 3 *(Z) — decoder radius-completeness violations established exhaustively, without syndrome-noise contamination;* (ii) *under frozen noisy replay, BP+OSD preserves the certified boundary (max wrong-logical LFR over $W \leq 2$ controls* 0.16–1.96% *vs certified halves* 73–87%*), whereas UF and BP-guided MWPM show near-zero wrong-logical rates* because they abstain*: 44–58% of their noisy $W \leq 2$ corrections carry a nonzero residual syndrome (category* `invalid_residual`*;* 71–98% *at weight 2 alone), i.e. they fail to produce any syndrome-consistent correction rather than a wrong one (Table 3).* (iii) Corrected: *the v23 statement that UF and BP-guided MWPM attain $\geq$ 99% logical failure at weight 2 conflated invalid corrections with logical errors under the old endpoint; under the four-category endpoint*

their wrong-logical rates at $W \leq 2$ are $\leq 0.4\%$ (noisy) with the exhaustive noiseless violations of (i) the correct statement of decoder instability. Radius-completeness of the decoder is therefore a security *property, not only a performance property: the adversarial work factor $t{+}1 = 3$ of Corollary 4 is realised by BP+OSD but is not inherited by non-radius-complete graph decoders, whose failures surface as invalid (residual-carrying) corrections rather than syndrome-consistent wrong ones.*

Table 3: Injection-boundary decoders, v24 frozen paired protocol (BB72, single-round phenomenological $p = 0.003$, frozen probe set and frozen native-error replay, four-category endpoint of Definition 3). "Noiseless $w{\leq}2$": exhaustive wrong-logical counts over all 2,629 masks per sector (no syndrome noise). "Controls": noisy $W \leq 2$ injections (13 probes, 8,000 trials each); "halves": the 8 O5-certified weight-3 halves of weight-6 logicals (4,000 trials each); WL = valid wrong-logical rate; inv. = share of corrections whose residual has nonzero *true* syndrome (mean over the 13 control probes). Endpoint validity is tested on the true residual syndrome (second-round-review repair; regression tests T1–T4 run at every launch of the runner); the archived v24 arrays contain zero trials of the mislabel type the review exhibited, 2.24% of trials (true-syndrome-consistent corrections previously counted invalid) are reclassified, and all values here regenerate from the repaired runner. Files: `data/inj_radius_noiseless_v24.csv`, `data/injection_boundary_decoders_v24.csv`, `data/injection_boundary_mcnemar_v24.csv`.

	noiseless $w{\leq}2$ WL		controls: max WL		controls: inv. share	
decoder	X	Z	X	Z	X	Z
BP+OSD	0	0	0.0204	0.0016	0.044	0.046
MW-lookup	0	0	0.0173	0.0004	0[a]	0[a]
UF	13	0	0.0040	0.0001	0.510	0.569
BP-guided	0	3	0.0006	0.0004	0.416	0.430

[a] MW-lookup aborts on 11% of control trials (aborts excluded from WL). Certified halves (weight 3): WL range 0.12–0.90 (BP+OSD, X), 0.89–0.91 (BP+OSD, Z), 0.008–0.741 (MW-lookup), ≤ 0.018 (UF), ≤ 0.044 (BP-guided). *Availability:* the restoration-failure rate (1 − valid-correct share; columns `restoration_failure`, `rf_*` of `injection_boundary_decoders_v24.csv`) has control-probe maxima 0.070 (BP+OSD), 0.136 (MW-lookup), 0.974 (BP-guided), 0.994 (UF) — an always-aborting decoder scores 1, not 0, under this measure.

Proposition 1 (Pauli-channel diamond bridge — corrected)**.** *Let Λ be a Pauli channel on one logical qubit with distribution $\{p_P\}$. Then $\frac{1}{2}\|\Lambda - \mathrm{id}\|_\diamond = 1 - p_I$. Writing the per-sector logical failure rates $\mathrm{LFR}_Z = p_X + p_Y$ (an X- or Y-type logical error fails the protected Z observable) and $\mathrm{LFR}_X = p_Z + p_Y$, we have*

$$\max_{\beta\in\{X,Z\}} \mathrm{LFR}_\beta \;\leq\; 1 - p_I \;=\; \mathrm{LFR}_{\mathrm{either}} \;\leq\; \mathrm{LFR}_X + \mathrm{LFR}_Z, \tag{3}$$

where $\mathrm{LFR}_{\mathrm{either}} = \Pr[\text{at least one sector fails}] = 1 - p_I$. Hence, if the effective logical channel of the implemented memory is Pauli (e.g. after Pauli twirling of the recovery) and the endpoint counts a trial as failed when either sector logical-fails*, then exactly $\varepsilon_{\mathrm{FT}}^{\mathrm{logical}} = 2(1 - p_I) = 2\,\mathrm{LFR}_{\mathrm{either}}$; with a per-sector (*max*) endpoint only the two-sided estimate $2\max_\beta \mathrm{LFR}_\beta \leq \varepsilon_{\mathrm{FT}}^{\mathrm{logical}} \leq 2(\mathrm{LFR}_X + \mathrm{LFR}_Z)$ is available. The v23 version of this proposition used $2\max_\beta \mathrm{LFR}_\beta$ as an* upper *bound; that inequality is false (counterexample: $p_X = p_Z = 0.1$, $p_I = 0.8$ gives $2(1 - p_I) = 0.4 > 0.2 = 2\max_\beta \mathrm{LFR}_\beta$) and is corrected here by the either-sector formulation.*

Proof. Pauli channels are covariant; the diamond norm is attained on a maximally entangled input and equals the ℓ_1 distance of the Pauli distributions [43]: $\sum_P |p_P - \delta_{P,I}| = 2(1 - p_I)$. The sector identities $\mathrm{LFR}_Z = p_X + p_Y$ and $\mathrm{LFR}_X = p_Z + p_Y$ are immediate from the CSS structure, and the chain of inequalities follows from $p_Y \geq 0$ and $p_X + p_Y + p_Z = 1 - p_I$. □

Remark 8 (Instantiated bound for BB72)**.** *The single-round experiment counts a trial as failed if either sector logical-fails, so under the Pauli approximation Proposition 1 applies with equality:* $\varepsilon_{\mathrm{FT}}^{\mathrm{logical}} = 2\,\mathrm{LFR}_{\mathrm{either}}$*. For BB72 with BP+OSD at* $p = 0.004$*,* 160 *failures in* 10^6 *trials (LFR* 1.60×10^{-4}*, Wilson 95% CI* $[1.37, 1.87] \times 10^{-4}$*, Table 8) give* $\varepsilon_{\mathrm{FT}}^{\mathrm{logical}} = 3.2 \times 10^{-4}$ *as a point estimate, and* $\leq 3.74 \times 10^{-4}$ *at 95% confidence, under the Pauli approximation.*

Remark 9 (Calibrated twirling residual — analytic closure)**.** *The Pauli restriction of Proposition 1 is quantified exactly for the canonical REP3 bit-flip memory under coherent axis-aligned noise* $U^{\otimes 3}$*,* $U = e^{-i\theta X}$*, with ideal extraction and majority-vote recovery (`run_twirling_residual.py`, `run_twirling_analytic_v24.py`, `data/twirling_residual_v24.csv`). Writing* $s = \sin\theta$*,* $c = \cos\theta$*, the logical channel has logical flip probability* $p_L = 3s^4c^2 + s^6$ *and coherent off-diagonal PTM element* $g = 2c^3s^3$*; the channel difference from its Pauli twirl is unitarily equivalent to a Z-rotation, so the diamond distances are attained on basis inputs and are* exact *[43]:*

$$\|\mathcal{E}_L - \mathcal{E}_L^{\mathrm{twirl}}\|_\diamond = 2|g| \approx 4\theta^3, \qquad \|\mathcal{E}_L - \mathrm{id}\|_\diamond = 2\sqrt{p_L^2 + g^2}. \tag{4}$$

The archived variational values of v23 (`data/twirling_residual.csv`) coincide with these exact expressions at all eight grid points up to archived rounding ($2|g|$*:* 3.2×10^{-5} *at* $\theta = 0.02$ *to* 9.0×10^{-2} *at* $\theta = 0.30$*), upgrading the v23 lower estimates to closed-form equalities. Three readouts: (i) the logical failure rate is twirl-exact,* $\mathrm{LFR}_{\mathrm{coh}}/\mathrm{LFR}_{\mathrm{twirl}} = 1.000$ *for all* $\theta \leq 0.3$ *— axis-aligned coherence changes the logical channel's phase, not its flip statistics; (ii) the residual logical rotation is cubic,* $\theta_L \approx -2\theta^3$*; and (iii) the bridge ratio* $\|\mathcal{E}_L - \mathrm{id}\|_\diamond/(2\,\mathrm{LFR}) = \sqrt{1 + (g/p_L)^2} \approx 2/(3\theta)$ *grows without bound as* $\theta \to 0$*, from* 33.3 *at* $\theta = 0.02$ *to* 2.3 *at* $\theta = 0.3$ *— outside the Pauli class no* θ*-independent bridge exists, while within it Proposition 1 is exact. This is a regression example in the sense of Huang, Doherty, and Flammia [36]: coherent errors that are invisible to LFR alone inflate the diamond distance. The instantiated bridge numbers of Remark 8 therefore carry a quantified, not merely acknowledged, Pauli caveat.*

Corollary 3 (Device-aware stochastic-injection bound)**.** *For i.i.d. injection at rate* p *per data qubit in one round, with* n_d *data qubits in the protected sector,*

$$\lambda(p) \;\leq\; \sum_{w=0}^{n_d} \Pr[\mathrm{Bin}(n_d, p) = w]\, \pi((t{+}1) - w),$$

from $|N \oplus F| \leq |N| + |F|$ *and Theorem 3(c). For the repetition memory* $n_d = d$*; for a CSS memory the Binomial ranges over the* n_d *data qubits of the sector, not the distance (this corrects the v23 statement, which wrote* $\mathrm{Bin}(d, p)$ *without the sector scope). The bound uses only fault counts; weight-*w *patterns are mostly benign at small* w*, so it is a count-based upper bound, not a characterization.*

Remark 10 (Independence-form bound vs. composed model: theory and campaign chronology)**.** *The form* $\lambda(p) \leq \delta_0 + \Pr[\mathrm{Bin}(n_d, p) \geq t{+}1]$ *(Table 4) assumes a weight-*$\leq t$ *injection leaves the failure probability at baseline. Theorem 3(c) shows the two are* different models *whenever* $\pi(1) > 0$*: a weight-*w *injection needs only* $t{+}1{-}w$ *assisting native faults, so the composed bound can exceed the independence-form bound; whether the independence form fails on a device is an empirical question, not a consequence of the theorem.* Campaign chronology (corrected)*. The v23 text reported that the pinned campaign of Appendix B (predictions, decision rules, and masks hash-pinned before submission) falsified the independence form at* $p = 0.10$ *by* 2.1–3.8× *on both sectors of ibm_marrakesh. That verdict compared the observed rates against a fresh-mask threshold, while the campaign injected* 50 pinned *masks per cell. The mask-conditioned estimand that resolves this was adopted* retrospectively*, at reanalysis after inspection of the realized masks — it is not part of the pre-submission pinning (the full chronology is Table 23). Under that estimand (`theory_test_consol/run_stoch_reanalysis_v24.py`,*

*`results/stoch_reanalysis_v24.csv`), no stochastic cell exceeds the corrected naive threshold (e.g. at $p = 0.10$: observed $0.0211/0.0388$ (campaign 1) and $0.0202/0.0403$ (consolidation A) vs conditioned thresholds $0.0224/0.0427$), so the falsification claim is **withdrawn**. The campaign-1 deviations* motivated *the composed Binomial⊗Poisson model — a discovery, not a prespecified prediction; the confirmation chronology is stated in Appendix B.1, and a confirmatory experiment with freshly sampled masks, pinned in advance, is required to separate the two models on device.*

Proposition 2 (Independence-form i.i.d. bound — superseded on device)**.** *For adversary restricted to i.i.d. Bernoulli attacks at rate p_{atk} and implementation error δ_{FT} independent of the attack, $V^\star|_{\text{iid}} \geq (1-\delta_{\text{FT}})\Pr[\text{Bin}(n, p_{\text{atk}}) \leq t]$.* The independence assumption is the content: *on real hardware the implementation error is* not *independent of the attack — Theorem 3(c) gives the correct composition. The v23 claim that the pinned device test falsified this independence form at $p = 0.10$ is withdrawn under the mask-conditioned reanalysis (Remark 10). Table 4 evaluates the independence form for $n = 128$, $d = 11$, $t = 5$, $\delta_{\text{FT}} = 0.1$; use Corollary 3 for device-level predictions.*

Table 4: Independence-form i.i.d. stochastic fidelity lower bound ($n = 128$,$d = 11$,$t = 5$,$\delta = 0.1$): $0.9 \cdot \Pr[\text{Bin}(128, p) \leq 5]$. Restricted to the i.i.d. subclass *with attack-independent implementation error*; its device-level falsification claim of v23 is withdrawn (Remark 10) — the independence form is a different model, and Corollary 3 is the device-conditional bound.

p	LB	p	LB	p	LB
0.001	0.900	0.020	0.860	0.050	0.341
0.005	0.900	0.030	0.731	0.080	0.046
0.010	0.898	0.040	0.534	0.100	0.008

Remark 11 (Decoder-dependent ACE value — empirical observation)**.** *Theorem 2 establishes $V^\star \geq 1 - \varepsilon_{\text{FT}}$ for any implemented recovery; because ε_{FT} is decoder-dependent, different recoveries yield different ACE values. At $p = 0.008$, BB72 BP+OSD achieves LFR $= 0.0014$ while MWPM-DEM achieves LFR $= 0.1894$. Taking LFR as a proxy for ε_{FT}, the observed decoder-dependent difference is $0.1894 - 0.0014 = 0.1880$ at this operating point (the v23 text reported 0.191; corrected arithmetic).* Bound status: *under the Pauli approximation, Proposition 1 bounds each decoder's own logical floor by twice its either-sector LFR (2×0.0014 resp. 2×0.1894); a* certified gap between decoders *would require a lower bound on one side, which we do not establish, and the v23 inference of a certified gap of up to 0.379 is withdrawn. The coherent (non-Pauli) twirling residual is quantified separately in Remark 9.*

Proposition 3 (Syndrome-oracle insufficiency)**.** *Let an* integrity predicate *be any decision rule whose input is syndrome event statistics alone, and let the integrity quantity be the logical failure rate $\lambda_{\mathcal{M}}$ of the injection-budget game. Then syndrome monitoring is a strictly weaker certificate of memory integrity than logical-level injection probing: there exist configurations, on the same device and code family, whose syndrome statistics order them one way while $\lambda_{\mathcal{M}}$ orders them the other.*

Proof. (i) *Non-identifiability pair (explicit budget condition).* Let L be a nontrivial logical Pauli of the protected sector of weight $w(L)$, and split its support as $L = E_1 \oplus E_2$ with $|E_1| = \lceil w(L)/2 \rceil$, $|E_2| = \lfloor w(L)/2 \rfloor$; write σ for their common syndrome. Under deterministic minimum-weight decoding with a fixed tie-breaking rule, let $C(\sigma)$ be the decoder's correction, $|C(\sigma)| \leq |E_2|$, and write the residual cosets of the two half-attacks as $r_i = [C(\sigma) \oplus E_i]$. Since $r_2 = r_1 + [L]$ and $[L]$ is nontrivial, $r_1 \neq r_2$, so at least one half fails with certainty: writing $q(F)$ for the per-attack risk of injected pattern F (Remark 1), $\max(q(E_1), q(E_2)) = 1$, while E_1 and E_2

induce *identical* syndrome-event statistics. If the tie-break lands on one of the halves up to a stabilizer — always the case for single-logical ($k = 1$) codes, where $r_1 \in \{0, [L]\}$ — exactly one half is benign and (E_1, E_2) is a syndrome-indistinguishable pair with risks $(0, 1)$. For multi-logical codes the tie-break may return a different coset, and then *both* halves fail: in the $[[4, 2, 2]]$ code (stabilizers XXXX, $ZZZZ$) the split $L = X_1X_2$ gives the syndrome-identical faults X_1 and X_2, and a decoder answering X_3 leaves the nontrivial residuals X_1X_3 and X_2X_3 respectively. The earlier phrasing "exactly one half of any selected minimum logical is corrected" is therefore false in general and is withdrawn; the correct statement is per-attack. Whenever both halves fail, the pair $(C(\sigma), E_1)$ — the decoder's own correction, benign by construction, against a half that fails with certainty — is again a syndrome-indistinguishable $(0, 1)$ pair within budget, since $|C(\sigma)| \leq \lfloor w(L)/2 \rfloor$. Hence in all cases, for any injection budget $W \geq \lceil w(L)/2 \rceil$, the attack set contains two configurations with identical syndrome-event statistics and per-attack risks 0 and 1 respectively, so no predicate of syndrome statistics alone can separate a $\lambda_{\mathcal{M}} = 0$ configuration from a $\lambda_{\mathcal{M}} = 1$ configuration inside the same budget class; which member fails is decided by the decoder's tie-breaking, not by anything visible at the syndrome level. For BB72 the certified weight-6 logicals give such pairs already at $W = 3$ (Section 10), realized decoder-dependently in the injection studies of Appendix B. (ii) *Syndrome disturbance is not evidence of logical damage.* For $W \leq t$, Theorem 3(a) gives $\lambda_{\mathcal{M}}(W) = 0$ under ideal extraction, while each injected Pauli anti-commutes with its adjacent checks in every round, so the syndrome pattern is deterministically disturbed. (iii) *Pilot empirical illustration (exploratory; Appendix A).* On `ibm_marrakesh` REP5, the defect-avoiding B line runs strictly *hotter* at the syndrome level (per-detector event rates higher, $z = +3.69$ and $z = +3.87$ in the two bases) yet *colder* logically ($0/10^4$ vs $1/10^4$ in Z, $4/10^4$ vs $7/10^4$ in X). The pooled association is method-sensitive: CMH one-sided $p \approx 0.042$, two-sided ≈ 0.084, exact stratified ≈ 0.0517, odds ratio ≈ 1.43 with confidence interval spanning 1; we therefore report it as an exploratory pilot observation consistent with the non-identifiability structure, not as evidence of an ordering inversion. Since $\lambda_{\mathcal{M}}$ is directly measurable by logical-level injection (Definition 4), certification of integrity must include injection probing (Section 11.1). □

Lemma 1 (Logical-operator weight scaling)**.** *At encoding rate R in an independent surface-code patch architecture (d_i^2 qubits per patch), $\min_i d_i \leq \lfloor 1/\sqrt{R} \rfloor$. A qLDPC family with $d_n \geq \delta_0 n$ requires weight $\geq d_n = \Omega(n)$ for any nontrivial logical Pauli.* Elementary counting result; no overhead or latency accounting. *For $R = 1/6$: surface-code weakest patch $d \leq 2$; BB72 requires weight ≥ 6. The comparison concerns independent-patch architectures at matched rate only; it does not apply to concatenated, lattice-surgery-linked, or otherwise correlated multi-patch surface-code schemes, and it is not an overhead or latency comparison.* Rate delimitation: *BB72, BB144, and BB288 encode $k = 12$ logical qubits at rates $1/6$, $1/12$, and $1/24$ respectively; the matched-rate statement applies to BB72 alone, and the BB144/BB288 comparisons below are at strictly lower rates (no asymptotic rate claim is made for the BB family here).*

Corollary 4 (Adversarial work factor)**.** *Under the spacetime fault model with exact minimum-weight decoding — with* deterministic tie-breaking *among minimum-weight representatives, a* radius-complete *decoder (one that always returns a syndrome-consistent correction within the certified radius), and an adversary who* knows *code, schedule, and decoder (Definition 2) — the minimum adversarial weight that can force a logical failure is $t+1$, and the achieving pattern is explicitly constructible (Theorem 3(b)). A randomized decoder instead distributes the failure probability across the tied minimum-weight cosets: the per-attack risk $q(F)$ of each half of a certified logical is then a tie-breaking-weighted average over the coset representatives — not necessarily $1/2$ per half for multi-logical codes (Proposition 3(i)) — so the* forced-*failure threshold and the* failure-probability *threshold must be distinguished; we report the former here and the latter in the injection studies. At the rate-$1/6$ comparison point of Lemma 1, an adversary needs weight 1 against the weakest independent surface-code patch ($d \leq 2$) but weight 3 against*

BB72 ($d = 6$ certified exactly). At their own (lower) rates $1/12$ *and* $1/24$*, BB144 and BB288 raise the weight to* 6 *and* 9 *($d = 12, 18$ from the algebraic constructions); these are not matched-rate comparisons. Hence for any constant-rate code family with unbounded d_n the adversarial work factor grows without bound while the independent-patch baseline's remains* 1*; for a linear-distance family ($d_n \geq \delta_0 n$) the ratio is $\Omega(n)$.* Elementary consequence of Theorem 3(b) and Lemma 1; same scope restrictions as the lemma.

5 Single-round decoder comparison

5.1 Protocol

One syndrome cycle per trial; independent Bernoulli Pauli noise at rate p; noiseless syndrome measurement; 10,000 trials per point. All experiments use QEC Emulator `run_sweep` [1].

Code parameters. BB72: $(\ell, m) = (6, 6)$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$. BB144: $(\ell, m) = (12, 6)$, same polynomials. BB288: $(\ell, m) = (12, 12)$, $A = x^3 + y^2 + y^7$, $B = y^3 + x + x^2$ (Bravyi et al. Table 1 [6]).

BP+OSD. `BPOSDDecoder` [1, 9]: OSD-CS order 7; min-sum, scaling 0.625; 100 iterations; ldpc v2 API.

MWPM-DEM. `MWPMDecoder` [1, 10] with DEM chain decomposition and global boundary edges (Section 3.3). The MWPM-DEM arm is a deliberately naive reduction of the BB hypergraph to a chain graph: its role is to quantify the structural cost of ignoring hyperedge structure, not to represent best-practice matching decoding of BB codes. The appropriate-tool matching controls (Relay-BP and BP-augmented symmetry matching, which decode the same instances without naive chain reduction) are the pinned paired grid of Section 6.

Seeds and provenance. Seeds: `uint32`(SHA256(context_string)[0:4]) where context_string concatenates code, decoder, and p; stable across Python processes regardless of `PYTHONHASHSEED`. Every row in `single_round_all_codes.csv` includes failure count, seed, and SHA-256 digests of H_X and H_Z.

5.2 Results

Table 5 shows BP+OSD results; Table 6 shows MWPM-DEM results; Table 7 shows risk ratios, sharpened at $p = 0.004$ by the 10^6-trial rare-event campaign in Table 8.

Table 5: BP+OSD single-round, 10,000 trials, Wilson 95% CIs. Zero-failure rows report the exact one-sided 95% Clopper–Pearson UB $2.996/N$ (Remark 12). File: `single_round_all_codes.csv`.

Code	p	Failures	LFR	Wilson 95% CI
BB72	0.001	0	<0.000300	exact CP UB
BB72	0.002	0	<0.000300	exact CP UB
BB72	0.004	1	0.00010	[0.00002, 0.00057]
BB72	0.006	6	0.00060	[0.00028, 0.00131]
BB72	0.008	14	0.00140	[0.00083, 0.00235]
BB72	0.010	23	0.00230	[0.00153, 0.00345]
BB72	0.012	50	0.00500	[0.00379, 0.00659]
BB144	0.001–0.010	0 each	<0.000300	exact CP UB
BB144	0.012	6	0.00060	[0.00028, 0.00131]
BB288	0.001–0.012	0 each	<0.000300	exact CP UB (all 7 pts)

Table 6: MWPM-DEM single-round, 10,000 trials, Wilson 95% CIs. LFR increases with code size. File: `single_round_all_codes.csv`.

Code	p	Failures	LFR	Wilson 95% CI
BB72	0.001	46	0.0046	[0.00345, 0.00613]
BB72	0.004	595	0.0595	[0.05503, 0.06431]
BB72	0.008	1894	0.1894	[0.18184, 0.19720]
BB72	0.012	3392	0.3392	[0.32998, 0.34854]
BB144	0.001	170	0.0170	[0.01465, 0.01973]
BB144	0.004	1925	0.1925	[0.18489, 0.20035]
BB144	0.008	4988	0.4988	[0.48900, 0.50860]
BB144	0.012	7297	0.7297	[0.72091, 0.73832]
BB288	0.001	625	0.0625	[0.05792, 0.06741]
BB288	0.004	5059	0.5059	[0.49610, 0.51570]
BB288	0.008	8667	0.8667	[0.85990, 0.87322]
BB288	0.012	9669	0.9669	[0.96321, 0.97023]

Table 7: Katz 95% risk ratio (MWPM-DEM / BP+OSD), all $\chi^2\, p < 0.0001$. File: `single_round_all_codes.csv`.

Code	p	BPOSD	MWPM	RR (95% CI)
BB72	0.004	1	595	595× [83.7, 4230]
BB72	0.008	14	1894	135.3× [80.0, 228.7]
BB72	0.012	50	3392	67.8× [51.4, 89.6]
BB144	0.004	0	1925	∞ (BPOSD $k = 0$)
BB144	0.008	0	4988	∞ (BPOSD $k = 0$)
BB144	0.012	6	7297	>1,000× (Katz CI: [547, 2707])
BB288	0.004	0	5059	∞ (BPOSD $k = 0$)
BB288	0.008	0	8667	∞ (BPOSD $k = 0$)
BB288	0.012	0	9669	∞ (BPOSD $k = 0$)

Table 8: Rare-event refinement (E1): BP+OSD single-round logical failures in 10^6 trials per point. Nonzero rows: Wilson 95% CIs. Zero-failure rows: *exact* one-sided 95% Clopper–Pearson upper bound $1 - 0.05^{1/N} = 2.996 \times 10^{-6}$ (the Wilson score bound 2.706×10^{-6} is an approximation that is slightly anti-conservative at k=0; we report the exact value). Risk ratios vs MWPM-DEM (10,000 trials, Table 6) use the log method; the BB288 ratio combines the MWPM-DEM 97.5% one-sided lower bound with the BP+OSD exact 97.5% one-sided upper bound, giving joint confidence $\geq$ 95% by Bonferroni. File: `rare_event_bposd.csv`; seeds `stable_seed('E1', code, p)`.

Code	p	Failures	LFR	Wilson 95% CI	RR vs MWPM-DEM (95% CI)
BB72	0.002	22	2.20×10^{-5}	$[1.45, 3.33] \times 10^{-5}$	n/a (MWPM grid gap)
BB72	0.004	160	1.60×10^{-4}	$[1.37, 1.87] \times 10^{-4}$	371.9× [312.7, 442.3]
BB144	0.002	1	1.00×10^{-6}	$[0.02, 0.57] \times 10^{-5}$	n/a (MWPM grid gap)
BB144	0.004	11	1.10×10^{-5}	$[0.61, 1.97] \times 10^{-5}$	17,500× [9,678, 31,643]
BB288	0.002	0	$< 2.996 \times 10^{-6}$	exact one-sided UB	n/a (MWPM grid gap)
BB288	0.004	0	$< 2.996 \times 10^{-6}$	exact one-sided UB	>134,000× (joint $\geq$ 95%)

Table 8 sharpens the $p = 0.004$ column of Table 7: the 10^6-trial campaign replaces the single-failure and zero-failure counts with bounded estimates, resolving the BB144 and BB288 infinite risk ratios at this operating point.

Remark 12 (Zero-failure bounds and joint confidence)**.** *Throughout, a row with k=0 failures in N trials reports the exact one-sided Clopper–Pearson bound $1 - \alpha^{1/N} \approx 2.996/N$ at α=0.05 — never the Wilson score value $\approx 2.706/N$, which is anti-conservative at k=0. Risk ratios that combine a numerator interval with a zero-failure denominator interval are constructed at 97.5% per side, so the ratio bound holds with joint confidence $\geq$ 95% (Bonferroni). All campaigns reported here ran under pinned stopping rules (fixed N or the pinned adaptive ladder of Section 7), so fixed-N exact bounds apply; for the adaptive ladder we additionally report the Howard–Ramdas time-uniform confidence sequences [40] alongside the fixed-N bounds, so that the stopping-time interpretation is anytime-valid (Section 7).*

5.3 Union-Find decoder: second graphified baseline comparison

To test whether the poor graph-decoder performance is specific to minimum-weight matching on the chain graphification, we implemented a Union-Find BFS decoder as a second graphified reference baseline. UF-BFS grows clusters from active syndrome nodes via breadth-first search and matches them greedily. Like DEM-MWPM, it reduces weight-3 syndrome hyperedges to graph edges; it is therefore not hyperedge-aware and remains subject to the same information-loss limitation. The comparison distinguishes two graphified decoding rules under the same submitted representation; it does not rule out future correlated or hypergraph-aware matching decoders.

Table 9 shows that UF-BFS performs *worse* than MWPM-DEM at all tested error rates. At $p = 0.008$: UF-BFS produces 1,903/10,000 failures (LFR 19%) vs MWPM-DEM at 1,894/10,000 (LFR 18.9%) vs BP+OSD at 14/10,000 (LFR 0.14%).

Table 9: UF-BFS vs MWPM-DEM vs BP+OSD, BB72, single-round, 10,000 trials. File: `single_round_uf_decoder.csv`.

Decoder	$p = 0.002$	$p = 0.004$	$p = 0.008$	$p = 0.012$
BP+OSD	0/10k (<0.000300)	1/10k (0.00010)	14/10k (0.00140)	50/10k (0.00500)
MWPM-DEM	151/10k (0.0151)	595/10k (0.0595)	1894/10k (0.1894)	3392/10k (0.3392)
UF-BFS	182/10k (0.0182)	614/10k (0.0614)	1903/10k (0.1903)	3623/10k (0.3623)

Scientific interpretation. Both tested graphified baselines fail substantially more often than BP+OSD under this implementation. A plausible mechanism is the BB Tanner graph's weight-3 column structure: every data-qubit error can trigger three syndrome checks simultaneously, creating hyperedges that the submitted graphifications reduce to pairwise edges with information loss. BP+OSD operates directly on the Tanner graph and handles weight-3 variable nodes natively. The consistent three-way ordering BP+OSD $\ll$ MWPM-DEM $<$ UF-BFS suggests that the BP+OSD advantage is consistent across the two graphified reference baselines tested here; it does not prove that all conceivable MWPM-class implementations fail. File: `single_round_uf_decoder.csv`.

6 Modern decoder comparator: a pinned paired grid

6.1 Protocol: pairing, pinning, and certification

The single-round comparison of Section 5 predates the 2026 decoder literature. We therefore re-ran the BB72 single-round point against six additional decoders spanning the three post-2024 families, under a deliberately stricter statistical protocol:

- **Paired error instances.** Every decoder decodes the same 10,000 sampled error instances per rate p (seed `stable_seed("COMPARE","BB72",p)`; the vectorized generator was verified bit-identical to the archived scalar stream). Pairing enables exact McNemar discordant-pair tests between decoder pairs rather than overlapping unpaired confidence intervals.
- **Certified code instance.** The symatch authors' BB72 construction was verified matrix-identical to our archived matrices under an explicit column permutation (rows identical), recorded in `bb72_symatch_equiv_cert.json`; all comparisons therefore concern one and the same code.
- **Decoders.** BP+OSD (archive settings, Sec. 5); Relay-BP [26] ($\gamma_0 = 0.1$, 60 pre-iterations, 60 relay sets, $\gamma \in (-0.24, 0.66)$, stop-after-5); `symatch` [28] in five configurations: bare symmetry matching, over-matching ("simplex") on all $2^6 - 1$ symmetry combinations, with optional predecoding, and two BP-augmented variants (radius-2 exact: 2556/2556 weight-2 errors corrected).

A trial is a failure if either sector (X or Z) logical-fails. Artifacts: `comparator_grid.csv`, `comparator_mcnemar.csv`, fail-bit arrays per (decoder, p) in `results/failbits/`.

6.2 Results

Table 10: Pinned paired BB72 grid: logical failures per 10,000 paired trials (either sector). Columns: BP+OSD; Relay-BP; symatch BP+simplex (BPs) and BP+pre+simplex (BPps); simplex (sx); pre+simplex (psx); bare. File: `comparator_grid.csv`.

p	BP+OSD	Relay	BPs	BPps	sx	psx	bare
0.001	0	0	0	0	7	4	367
0.002	1	1	1	1	25	16	760
0.004	1	0	0	0	117	63	1630
0.006	8	6	6	6	298	199	2360
0.008	12	14	13	13	458	324	2950
0.010	35	33	31	31	765	594	3595
0.012	43	40	33	33	1062	836	4349

Table 10 and Figure 2 support four claims. **(i) Bare symmetry matching is not viable on this BB72 instance**: 367/10,000 failures already at $p = 0.001$ (versus zero for all augmented variants), rising to 43.5% at $p = 0.012$; paired McNemar tests against every augmented variant give z up to 65.7. The mechanism is the weight-1 straddling failure documented in [28]: decoding all 72 single-qubit faults per sector through the archived comparator face (`comparator/run_symatch_w1_diagnostic_v24.py`, `results/symatch_w1_diagnostic_v24.json`) leaves 24/72 (X-sector) and 15/72 (Z-sector) uncorrectable — a genuine sector asymmetry, not an implementation artifact (BP-augmented variants correct 72/72 in both sectors). *Correction:* the v23 text reported 48/72 and 57/72 here; those counts were wrong and are replaced by the recomputed values, which agree with the independent review diagnostic. **(ii) BP augmentation closes the gap within an equivalence margin adopted retrospectively**: taking ±0.5% absolute paired LFR difference as the equivalence margin (adopted for the confirmatory protocol and applied *retrospectively* to the archived grid; no pre-registration of this margin exists for the archived analysis — see the chronology in Appendix C), symatch-BP+simplex versus BP+OSD at the largest rate $p = 0.012$ differs by 10 vs 20 discordant pairs, i.e. an absolute paired difference of 0.10% (95% CI $[-0.01\%, 0.21\%]$) — inside the margin; the same holds at every rate, and BP+OSD versus Relay-BP is inside the margin everywhere (smallest exact McNemar $p = 0.69$). We report equivalence at this margin, not point-null equality. *Setting disclosure:* the symatch matching graphs were built with a fixed $p_{\text{phys}} = 0.001$ prior at all grid rates (a deliberately mismatched prior off $p = 0.001$); a matched-prior control with a documented equal tuning budget is assigned to the confirmatory phase, and all matching conclusions here are limited to the tested construction. **(iii) Predecoding is redundant under BP**: the BP+pre+simplex and BP+simplex configurations agree on all 70,000 paired instances (zero discordant pairs at every rate), whereas without BP the predecode step is worth a significant factor (836 vs 1062 failures at $p = 0.012$, $z = -15.0$). **(iv) Symmetry exploitation alone carries a residual cost**: bare over-matching trails the BP-augmented variants by ≈32.2× at the highest rate (1062 vs 33 failures at $p = 0.012$; v23 mis-stated this ratio as "2–3×"), so on this code the six-symmetry decomposition functions best as a posterior-consistent fallback for BP — consistent with the augmentation requirement reported in [28] itself. Taken together, on this instance, endpoint, and tested set of constructions, the performance gap between graphified and native-graph decoders is captured by the presence or absence of BP on the Tanner graph, not by the choice of graphification rule; we do not claim an intrinsic impossibility for any tool family beyond the tested constructions.

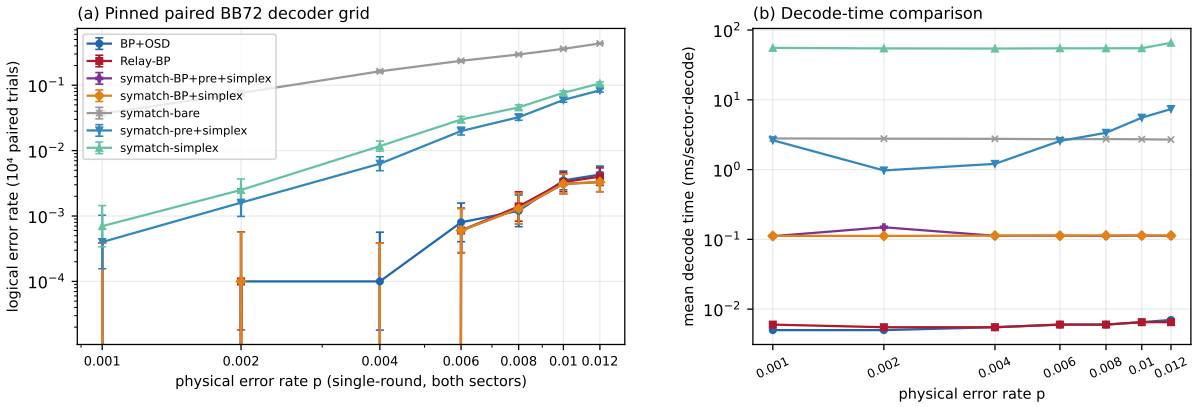


Figure 2: (a) Pinned paired BB72 decoder grid, Wilson 95% CIs; (b) decode time per sector-decode (single-process probe). The matching-core decoders are 10^2–10^4× slower per decode than the BP family while being up to 10^4× less accurate (bare) or statistically tied (BP-augmented).

7 Repeated-cycle BP+OSD benchmark

7.1 Protocol

All experiments use QEC Emulator `run_repeated_cycle` [1]. d syndrome rounds per trial: $d = 6$ (BB72), $d = 12$ (BB144), $d = 18$ (BB288). Rounds 1–$(d-1)$: Bernoulli noise at rate p on data and measurement bits. Round d: noiseless. Space-time extended check matrix per Eq. (1). Trials: BB72/BB144 at 2,000; BB288 at 500.

7.2 Results

Table 11: Repeated-cycle BP+OSD, d rounds per trial, Wilson 95% CIs (nonzero rows); zero-failure rows report the exact one-sided 95% Clopper–Pearson UB $2.996/N$ (Remark 12): $<$ 0.00150 (n = 2,000); $<$ 0.00599 (n = 500). v24 rerun with exactly aligned spacetime priors (`model_rev=v24-aligned-priors`; final-round fault variables removed, all surviving priors = p; the v23 grid, which assigned priors to nonexistent final-round faults, is superseded and archived). Files: `repeated_cycle_bposd_v24.csv`, `repeated_cycle_bb288_bposd_v24.csv`, `repeated_cycle_all_codes_v24.csv`.

Code (rounds, trials)	p	Failures	LFR	Wilson 95% CI
BB72 ($d = 6$, $n = 2000$)	0.001	0	<0.00150	exact CP UB
BB72 ($d = 6$, $n = 2000$)	0.002	0	<0.00150	exact CP UB
BB72 ($d = 6$, $n = 2000$)	0.004	1	0.00050	[0.00009, 0.00283]
BB72 ($d = 6$, $n = 2000$)	0.006	3	0.00150	[0.00051, 0.00440]
BB72 ($d = 6$, $n = 2000$)	0.008	10	0.00500	[0.00272, 0.00918]
BB72 ($d = 6$, $n = 2000$)	0.010	24	0.01200	[0.00808, 0.01779]
BB72 ($d = 6$, $n = 2000$)	0.012	37	0.01850	[0.01345, 0.02540]
BB144 ($d = 12$, $n = 2000$)	0.001–0.012	0 each	<0.00150	exact CP UB (all 7 pts)
BB288 ($d = 18$, $n = 500$)	0.001–0.012	0 each	<0.00599	exact CP UB (all 7 pts)

At $p = 0.008$ under d rounds: BB72 gives 10/2000 (LFR 0.50%, Wilson CI $[0.27\%, 0.92\%]$); BB144 gives 0/2,000 (exact CP UB $< 0.15\%$); BB288 gives 0/500 (exact CP UB $< 0.60\%$). With aligned priors the fixed grid is monotone within each code and BB144 shows no failures at any grid point; the v23 BB144 tail $(2, 1, 0$ at $p = 0.008, 0.010, 0.012)$ does not reproduce under the corrected priors and is superseded. Trial counts are fixed ex ante (2,000/500), with no adaptive stopping; the $p = 0.001$ and $p = 0.002$ points are new in v24 (the grid now spans 0.001–0.012 uniformly).

Table 12: Repeated-cycle adaptive extension (E3, v24 prior-aligned rerun): d rounds per trial, 2,000-trial batches, stopping at $\geq$ 50 failures or 20,000 trials; batch seeds `stable_seed('E3v24', code, p, rounds, batch)` (the $p = 0.010$ rows use tag `E3v24p010`). All rows: 95% Robbins uniform-mixture confidence sequences [40], time-uniform and valid under the pinned stopping rule (fixed-sample formulas would not be valid: nominal-95% Wilson coverage under this rule is $\approx$93.9% at $p = 0.021$, `stats_intervals.py -self-check`); zero rows report the CS upper endpoint 6.45×10^{-4} ($n = 20{,}000$). Files: `repeated_cycle_adaptive_*_v24.csv` (raw counts), `rc_adaptive_cs_v24.csv` (intervals).

Code (rounds)	p	Failures	Trials	LFR	95% confidence sequence
BB72 ($d = 6$)	0.004	16	20,000	8.0×10^{-4}	$[0.19, 2.10] \times 10^{-3}$
BB72 ($d = 6$)	0.006	60	20,000	3.0×10^{-3}	$[1.59, 5.07] \times 10^{-3}$
BB72 ($d = 6$)	0.008	62	10,000	6.2×10^{-3}	$[3.41, 10.2] \times 10^{-3}$
BB72 ($d = 6$)	0.010	64	4,000	1.60×10^{-2}	$[9.20, 25.5] \times 10^{-3}$
BB72 ($d = 6$)	0.012	84	4,000	2.10×10^{-2}	$[13.1, 31.5] \times 10^{-3}$
BB144 ($d = 12$)	0.004	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB144 ($d = 12$)	0.006	1	20,000	5.0×10^{-5}	$[\sim 0,\ 7.82] \times 10^{-4}$
BB144 ($d = 12$)	0.008	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB144 ($d = 12$)	0.010	2	20,000	1.0×10^{-4}	$[\sim 0,\ 8.99] \times 10^{-4}$
BB144 ($d = 12$)	0.012	4	20,000	2.0×10^{-4}	$[4.5 \times 10^{-6},\ 1.10 \times 10^{-3}]$
BB288 ($d = 18$)	0.004	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB288 ($d = 18$)	0.006	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB288 ($d = 18$)	0.008	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB288 ($d = 18$)	0.010	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint
BB288 ($d = 18$)	0.012	0	20,000	$<6.45 \times 10^{-4}$	CS upper endpoint

Three-code finite-size evidence. The single-round data (10,000 trials) provides complementary finite-size evidence across all three code sizes. The size ordering LFR(BB72) > LFR(BB144) $\geq$ LFR(BB288) holds at every tested p value (e.g., $p = 0.008$: 14/10,000 vs 0/10,000 vs 0/10,000; ratio BB72/BB288 infinite at $p = 0.008$, χ^2 $p < 0.0001$). This ordering is the expected signature of sub-threshold operation: larger-distance codes fail more slowly below threshold. Combined with the repeated-cycle data showing zero BB144 failures on the entire v24 fixed grid (0/2,000 at each of 7 points; 4/20,000 worst-case on the adaptive extension) and zero BB288 failures at d rounds per trial on both grids (0/500 fixed; 0/20,000 adaptive at every tested p), the three-code dataset constitutes finite-size evidence that BP+OSD operates in a sub-threshold regime for these code sizes at the tested error rates. A formal finite-size scaling analysis establishing the asymptotic threshold requires additional code sizes, a circuit-level noise model, and finite-size scaling fits, and is identified as future work.

Distance-ladder suppression factor Λ. The adaptive extension gives a three-rung distance ladder ($d = 6, 12, 18$) under one pinned protocol. *Definition and scope.* We report Λ as the cube root of the *per-block* failure ratio, $\mathrm{LFR}_d/\mathrm{LFR}_{d+6} = \Lambda^3$ (and $\mathrm{LFR}_6/\mathrm{LFR}_{18} = \Lambda^6$ for the two-step combination), where a block spans d syndrome rounds; Λ is therefore a per-Δd=2 suppression of block failure, *not* a per-cycle suppression factor. The surface-code reference value $\Lambda = 2.14 \pm 0.02$ [31] was measured on hardware at circuit level with per-round benchmarking; our protocol is *not* matched to it, and we claim no comparison of absolute hardware performance — only of qualitative sub-threshold suppression in a phenomenological model. *v24 numbers* (`lambda_ladder_v24.csv`). Every ladder arm is adaptively stopped, so all endpoints are confidence-sequence based (four-endpoint Bonferroni for two-sided ratios, two-endpoint for one-sided; `stats_intervals.py`): BB72→BB144 suppression is 60× observed with CS lower bound > 1.77× ($\Lambda = 3.91$ observed, $\Lambda > 1.21$) at $p = 0.006$; > 4.9× ($\Lambda > 1.70$) at $p = 0.008$;

160× observed, $> 9.0\times$ ($\Lambda = 5.43$, $\Lambda > 2.08$) at $p = 0.010$; and 105× observed, $> 10.6\times$ ($\Lambda = 4.72$, $\Lambda > 2.19$) at $p = 0.012$. At $p = 0.004$ the rung is *not* resolvable under sequential-valid intervals (the CS ratio lower bound is below 1); the v23-era claim of significant suppression at that rate does not survive the corrected inference and is withdrawn. Upper endpoints are uninformative where the higher-distance arm has $O(1)$ counts — we report the point estimate and the lower bound plainly. The BB144→BB288 rung is *statistically unresolvable* at 20,000 trials per point (0/20,000 BB288 failures at every tested p; 0/20,000 BB144 failures except 1 at $p = 0.006$, 2 at $p = 0.010$, and 4 at $p = 0.012$): the one-sided ratio bounds are vacuous there, and we report no point estimate for that rung. The combined $d = 6 \to 18$ comparison gives $> 18.9\times$ ($\Lambda > 1.63$ per Δd=2) at $p = 0.012$, $> 13.2\times$ ($\Lambda > 1.54$) at $p = 0.010$, and $> 4.9\times$ ($\Lambda > 1.30$) at $p = 0.008$. The v23 ladder numbers (35.3× etc.) were computed under the misprioritized v23 spacetime matrices and are superseded.

8 Circuit-level corroboration

The single-round and paired-grid results above use code-capacity noise. Two circuit-level arms test whether the decoder ordering survives contact with realistic detector error models (DEMs).

8.1 Arm A: third-party re-analysis of the Kunlun hardware DEM

What this arm is. A re-analysis of the *released detector error model*, not of raw hardware shots: the Kunlun collaboration published the complete detector check matrices and per-mechanism channel probabilities of their $[[18, 4, 4]]$ BB hardware experiment [32] (Zenodo, CC-BY-4.0), and all numbers below are Monte-Carlo samples drawn from that published DEM, decoded independently by us. No new hardware data are involved, and no claim about the Kunlun device beyond the authors' own released model is made. We sampled 10,000 shots per point from the authors' own DEM at 1, 3, and 6 QEC rounds, both sectors, and decoded with five decoders: BP+OSD (authors' configuration, 10^4 iterations, CS-7), BP-only, static-prior MWPM on the chain-graphified DEM, belief-matching [29] (BP posteriors reweight the matching graph per shot), and Relay-BP [26]. File: `kunlun_cmp.csv`; Figure 3.

Table 13: Five decoders on the published Kunlun $[[18, 4, 4]]$ DEM, 10,000 shots per point. Bold: best (or statistically tied-best) per row. Ordering Relay-BP ≤ BP+OSD < BP-only ≪ BeliefMatch < MWPM holds in all six configurations.

Rounds	Sector	Relay-BP	BP-OSD	BP-only	BeliefMatch	MWPM-prior
1	X	**0.2082**	0.2103	0.2324	0.4231	0.4270
1	Z	**0.2223**	0.2248	0.2502	0.5299	0.5365
3	X	**0.5489**	**0.5489**	0.5897	0.7507	0.7653
3	Z	**0.5123**	0.5190	0.5626	0.7836	0.8155
6	X	**0.7887**	0.7952	0.8208	0.8921	0.9026
6	Z	**0.7670**	0.7728	0.8111	0.9109	0.9237

Three conclusions. **(i)** The graphification penalty measured in Section 5 is not an artifact of our synthetic DEM: on the authors' hardware DEM, matching-family decoders are 2.0–2.4× worse than BP+OSD at round 1, the penalty decaying to 1.1–1.2× by round 6 as the LER saturates — early rounds carry the information. **(ii)** Per-shot belief reweighting recovers only 0.4–3.2 percentage points over the static-prior matching baseline, quantitatively mirroring the modest decoder-side gains of the defect-aware arm in [33] on independent hardware. **(iii)** Relay-BP matches or beats the 10^4-iteration BP+OSD reference at $\sim 35\times$ lower wall time per point (8 s vs 288 s average, same machine, single process, identical inputs), consistent with its real-time design goal [26, 27].

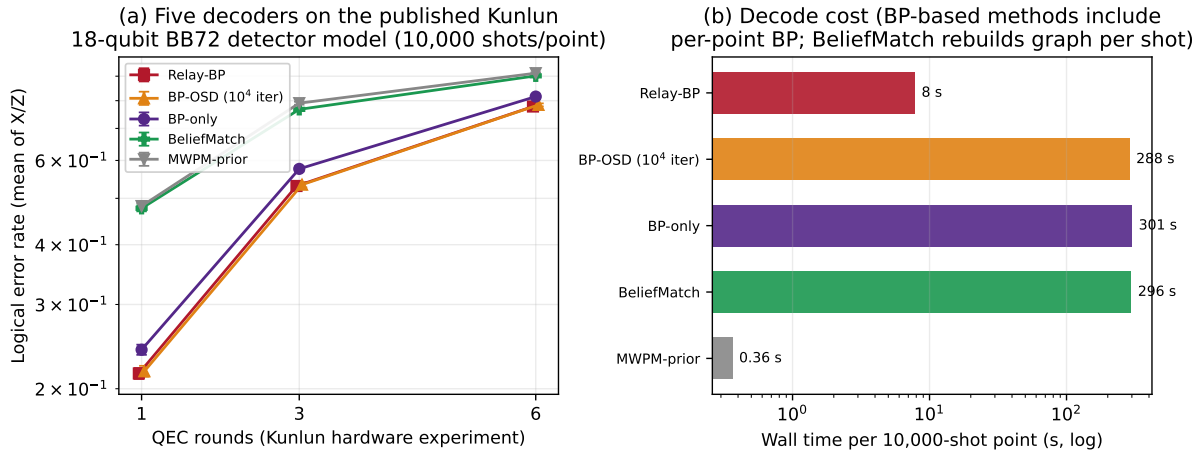


Figure 3: (a) Five decoders on the Kunlun hardware DEM, mean of X/Z sectors, Wilson 95% CIs; (b) wall time per 10,000-shot point.

8.2 Arm B: synthetic BB72 circuit-level point

We close the loop on the code-capacity graphification penalty of Section 5 with one synthetic circuit-level point: a BB72 Z-memory experiment (4 extraction rounds, DEPOLARIZE1/2 gate noise, measurement flips at rate p, data-idle noise during measurement windows, and *ideal* reset/preparation — no explicit reset-preparation error channel, disclosed; serial-by-check schedule) compiled with Stim [12], 20,000 shots per rate, decoded by BP+OSD, Relay-BP, belief-matching, and two matching baselines. *C01 repair.* The v23 version of this arm built both extraction layers with reversed CNOT directions, so its DEM did not represent BB72 syndrome extraction; the v23 grid is withdrawn (archived as `wsc_circuit_v23_reversed_extraction_INVALID.csv` with a withdrawal note). The corrected circuit (`circuit_rev=v24-corrected-extraction`) is pinned by four protocol-invariance test groups — gadget parity (the reversed primitive returns input-independent zero), codespace silence, injection-column alignment with H_Z/H_X, and a detector/observable map audit — all passing deterministically (`wsc_verify.py`, `wsc_verify_report.json`). The corrected campaign regenerates the grid rate by rate under a checkpointed, resume-safe protocol with per-point DEM digests; the $p = 0.001$ point below is complete and verified, and the remaining rates deposit to the archive as they land (`wsc_circuit_v24.csv`); Figure 4.

Table 14: Synthetic BB72 circuit-level Z-memory, 4 rounds, 20,000 shots, corrected extraction (C01 repair; the reversed-extraction v23 grid is withdrawn). Failures (LER); zero-failure rows report the exact one-sided 95% Clopper–Pearson UB (Remark 12). MWPM-stim failures are entirely coverage failures (see text).

p	BP+OSD	Relay-BP	BeliefMatch	MWPM-chain	MWPM-stim
0.001	51 (0.0026)	0 ($<1.50\times10^{-4}$)	12826 (0.6413)	12848 (0.6424)	13824 (0.6912)

Three findings, all *strengthened* by the corrected extraction. **(i) The naive-reduction penalty is catastrophic at circuit level**: at $p = 0.001$ the chain-graphified matching decoder fails 64.2% of shots versus 0.255% for BP+OSD — a 252× penalty, an order of magnitude larger than the withdrawn v23 estimate. **(ii) Relay-BP's advantage is a circuit-level phenomenon**: under code-capacity noise (Section 6) Relay-BP and BP+OSD are statistically tied, but on this circuit DEM Relay-BP fails zero of 20,000 shots (exact CP UB 1.50×10^{-4}) while BP+OSD sits at 2.55×10^{-3} (Wilson CI $[1.94, 3.35] \times 10^{-3}$, excluding the Relay-BP

UB) on identical shots — quantitatively consistent with the circuit-level superiority reported in [26] and, on this tested operating point, associating that superiority with the noise model rather than the code; we do not extrapolate a general decoder ordering from a single corrected circuit point. **(iii) The standard matching toolchain cannot even be instantiated natively**: Stim's hyperedge decomposition fails on this DEM outright (weight-6 checks induce correlated mechanisms that do not decompose); with `ignore_decomposition_failures` the dropped mechanisms leave 69.1% of shots without any perfect matching at $p = 0.001$ (every such shot is a counted failure, and all 13,824 MWPM-stim failures are of this form). Within the tested reductions and this pinned circuit configuration, weight-6-check qLDPC circuits therefore sit outside the representational reach of the pairwise-matching stack: both tested routes to a pairwise model (Stim's native hyperedge decomposition and the independent-merge chain graphification) either fail outright or discard the correlated mechanisms, and BP-family decoders are the only operable option among those tested. We do not claim that no bespoke pairwise reduction exists; a general inexpressiveness statement would require a formal argument beyond the tested constructions.

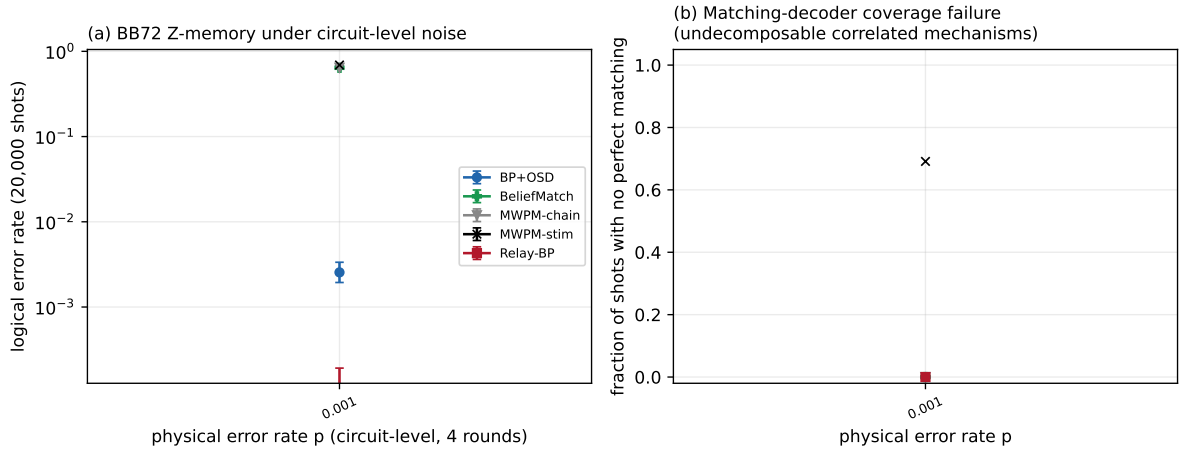


Figure 4: (a) BB72 circuit-level Z-memory LER, Wilson 95% CIs; (b) fraction of shots for which no perfect matching exists.

9 Hardware noise-model simulations with confounder ablations

Single-cycle BB72 simulations using QEC Emulator `HardwareModel` [1], 5,000 trials (Table 17). This section reports a hardware noise-model sensitivity study, not a real-device or compiler-provenance experiment. The ablation quantifies how the simulated BB72 LFR changes when SWAP routing, measurement noise, idle noise, and backend gate-decoherence parameters are modified one at a time. Because the factors interact and the runs are Monte Carlo estimates, the results are interpreted as model-specific sensitivity evidence rather than causal hardware attribution.

9.1 Ablation protocol

We isolate four confounders by enabling and disabling each independently in a controlled simulation at $p_{\text{base}} = 5 \times 10^{-4}$ and 3,000 trials per condition. The four confounders are: (1) *SWAP routing overhead*: additional two-qubit gate errors from nonlocal check routing in the planar CZ architecture (≈ 96 SWAPs out of 432 two-qubit gates per cycle); (2) *gate decoherence*: thermal relaxation and dephasing during the gate ($p_1 \approx 1.33 \times 10^{-3}$ per CZ gate, negligible for MS); (3) *idle noise*: data-qubit decoherence between gate layers ($p_{\text{idle}} = 0.05\, p_{\text{base}}$); (4) *measurement*

noise: ancilla readout errors ($p_{\text{meas}} = p_{\text{base}}$).

9.2 Ablation results

Table 15 presents the ablation results. Within this simulation model, the *CZ: full model* condition has LFR = 0.0100, while *CZ: no SWAP overhead* gives LFR = 0.0050, so the simulated SWAP component changes the observed LFR by 0.0050. Removing measurement noise gives LFR = 0.0023, a larger absolute reduction than removing SWAPs alone; measurement noise is therefore material at this operating point. The *CZ: gate decoherence only* condition gives zero failures in 3,000 trials (Wilson upper bound < 0.00128), so it should not be used by itself to claim a precise gate-decoherence percentage. The MS CZ-matched-gate condition raises MS LFR from 0.00233 to 0.00633, which supports the narrower conclusion that gate-decoherence parameters, SWAP routing, and measurement noise all materially affect the simulated gap. No real-hardware causal attribution is claimed.

Table 15: Hardware confounder ablation study for BB72, $p_{\text{base}} = 5 \times 10^{-4}$, 3,000 trials each, radius-2 lookup decoder. Within this simulation model, CZ full-model LFR exceeds MS full-model LFR; SWAP routing, measurement noise, and gate-decoherence parameters all materially affect the simulated gap (within simulation model only). File: `hardware_ablations.csv`.

Condition	Failures	LFR	Wilson 95% CI
Phenomenological baseline	3	0.0010	[0.00034, 0.00294]
CZ: full model	30	0.0100	[0.00701, 0.01424]
CZ: no SWAP overhead	15	0.0050	[0.00303, 0.00823]
CZ: no idle noise	30	0.0100	[0.00701, 0.01424]
CZ: no measurement noise	7	0.0023	[0.00113, 0.00481]
CZ: gate decoherence only	0	<0.000999	exact CP UB (Rem. 12)
MS: full model	7	0.0023	[0.00113, 0.00481]
MS: CZ-matched gate error	19	0.0063	[0.00406, 0.00987]

Sensitivity summary. The CZ/MS separation should be reported as a simulation-model sensitivity result. At $p_{\text{base}} = 5 \times 10^{-4}$, SWAP removal changes CZ LFR by 0.0050, measurement-noise removal changes CZ LFR by 0.00767, and CZ-matched gate parameters raise the MS LFR by 0.0040. These shifts are comparable in scale, so the experiment does not support a single-cause explanation. *Idle noise is not a measurable contributor here:* removing it leaves the CZ arm unchanged (30/3,000 failures with and without idle noise; LFR 0.0100 both arms), so any idle-noise attribution is withdrawn — the model gives no evidence for an idle-window contribution at this operating point. The defensible conclusion is that routing overhead, measurement readout, and backend gate-decoherence parameters all contribute materially under the specified model.

Reset (ancilla-preparation) sensitivity. A referee-faced attribution question is whether imperfect ancilla reset (thermalization) is an unexamined alternative to idle-window decoherence: at single-cycle detection resolution the two mechanisms have identical signatures. We therefore ran a reset-sensitivity Monte-Carlo in the Pauli-propagating `CircuitLevelModel` (which propagates preparation faults through the schedule; BB72, $p = 5 \times 10^{-4}$, 3,000 trials per condition, radius-2 lookup decoder, pinned seeds; Table 16). Reset noise is *material and monotone*: LFR = 0.0637 (no reset), 0.0827 (nominal), 0.1077 (reset $\times 2$). At matched nominal syndrome-noise rate, reset-only and readout-only conditions give 0.0900 vs 0.0670 (non-overlapping Wilson CIs): preparation faults are *not* logically equivalent to readout faults — phase-type preparation errors propagate onto data qubits through the schedule — so reset and idle/readout attribution is separable in principle, and the reset contribution is now bounded

rather than unexamined. (The no-readout condition sits marginally above the full model; the difference is within Monte-Carlo jitter at 3,000 trials with overlapping Wilson CIs, and conditions are intentionally unpaired, so we report raw counts without smoothing.) Note that these absolute LFRs exceed those of Table 15 because the circuit-level model propagates two-qubit gate faults through the full schedule (a strictly harsher model than the effective-mask ablation); only within-table comparisons are meaningful.

Table 16: Reset-sensitivity Monte-Carlo, BB72 circuit-level model, $p = 5 \times 10^{-4}$, 3,000 trials per condition, radius-2 lookup decoder. File: `reset_sensitivity.csv`.

Condition	Failures	LFR	Wilson 95% CI
circuit: full model	248	0.0827	[0.07334, 0.09306]
circuit: no reset noise	191	0.0637	[0.05548, 0.07297]
circuit: reset ×2	323	0.1077	[0.09707, 0.11926]
circuit: no readout	288	0.0960	[0.08597, 0.10706]
circuit: reset-only syn	270	0.0900	[0.08028, 0.10077]
circuit: readout-only syn	201	0.0670	[0.05860, 0.07651]

9.3 Compiled single-cycle LFR comparison

Table 17: Hardware noise-model single-cycle LFR for BB72, 5,000 trials, radius-2 lookup decoder. Non-overlapping Wilson 95% CIs at $p = 5 \times 10^{-3}$. File: `hardware_three_condition_comparison.csv`.

p	Pheno fail.	Pheno CI	CZ fail.	CZ CI	MS fail.	MS CI
5×10^{-4}	0	<0.00077	10	[0.00109, 0.00368]	0	<0.00077
1×10^{-3}	2	[0.00011, 0.00146]	15	[0.00182, 0.00494]	0	<0.00077
2×10^{-3}	1	[0.00004, 0.00113]	46	[0.00690, 0.01225]	8	[0.00081, 0.00315]
3×10^{-3}	14	[0.00167, 0.00469]	77	[0.01234, 0.01920]	6	[0.00055, 0.00262]
5×10^{-3}	70	[0.01110, 0.01765]	171	[0.02951, 0.03961]	52	[0.00794, 0.01361]

Non-monotone row disclosure. The phenomenological column $(0, 2, 1, 14, 70)$ is non-monotone between $p = 1 \times 10^{-3}$ and $p = 2 \times 10^{-3}$ (2 vs 1 failures). Both points lie well inside each other's Wilson CIs, the trial streams are unpinned between conditions, and no adaptive stopping was used in this fixed-grid campaign (all trial counts fixed ex ante; the separate E3 extension of Section 7 does use a pinned adaptive stopping rule, analysed with time-uniform confidence sequences); we report the raw counts rather than a smoothed or monotone-constrained curve.

10 Finite-instance certificate (BB72)

Constructed via QEC Emulator `BB72()` [1] with `run_distance_certificate` (MitM enumeration).

H_X SHA-256: `7ab0973bfd02e399d69728d26e67dbfa...`;
H_Z SHA-256: `267f345c09710ec0cfad1e49e0786b26...`

O1. $H_X H_Z^T = 0$ over $\mathbb{F}_2$. *Pass (exact).* **O2.** $k = 72 - 30 - 30 = 12$. *Pass.* **O3.** $d \geq 6$: MitM exhaustive through weight 5, zero witnesses (2,629 left + 62,269 right patterns). **O4.** $t = 2$ correctability: all 2,629 weight-≤ 2 patterns, zero failures. *Pass (exhaustive).* **O5.** $d = 6$ exactly: exhaustive weight-6 enumeration (meet-in-the-middle 3+3 over all $\binom{72}{3} = 59{,}640$ triples per sector; every weight-6 kernel vector splits into two disjoint triples of equal syndrome, so the enumeration is complete) exhibits $84 + 84$ nontrivial weight-6 logicals, each verified syndrome-zero, outside the opposite stabiliser row space, and anticommuting with at least

one opposite-sector weight-6 logical (2,592 anticommuting pairs in total; lexicographically first witnesses $(0, 1, 2, 3, 20, 22)$ and $(0, 3, 6, 12, 27, 30)$). Combined with O3 (no nontrivial logical of weight ≤ 5), this certifies $d = 6$ exactly, not merely $d \geq 6$. *Pass (exhaustive).* Archive: `data/bb72_certificate_report.json` (obligations O1–O5).

Verification architecture (v24). Four properties make the certificate independently checkable. *(i) Sectorwise sufficiency.* O4 enumerates $2{,}629 = \sum_{w \leq 2} \binom{72}{w}$ patterns *per CSS sector*. A mixed Pauli of total weight ≤ 2 has an X-part and a Z-part each of weight ≤ 2, and CSS decoding treats the two sectors independently (X errors are corrected by H_Z, Z errors by H_X; $Y = iXZ$ decomposes into both), so sectorwise exhaustiveness covers *all* mixed-Pauli patterns of total weight ≤ 2 with no separate mixed enumeration. The same argument covers the noiseless exhaustive rows of Table 3. *(ii) Independent trusted checker.* `verify_certificate.py` is self-contained (archive-local, NumPy-only, 334 lines) and re-derives every obligation from the pinned matrices rather than trusting the generator (QEC Emulator): it recomputes the SHA-256 digests of H_X, H_Z against the pinned values, re-runs the meet-in-the-middle enumeration, and individually re-checks every published witness; it exits nonzero on any mismatch. *(iii) Published witnesses.* The report JSON publishes the full witness schema — matrix digests, enumeration cardinalities, and all 84+84 weight-6 logical supports with their anticommuting partners — so a third party can verify the *upper bound* $d \leq 6$ in linear time per witness (syndrome-zero, outside the opposite row space, anticommutation) without re-running the exponential search. Witnesses alone do not certify exactness: $d = 6$ additionally requires the lower-bound obligation that no nontrivial logical of weight < 6 exists, which the trusted checker re-establishes by re-running the meet-in-the-middle enumeration. The artifact therefore pairs linear-time witness checks (upper bound) with an exponential but independently re-runnable enumeration (lower bound). *(iv) Syndrome-consistency assertions.* Every O4 verdict is recomputed from the pinned check matrices, and every O5 witness is re-verified syndrome-zero before counting; a single inconsistency fails the run. The certificate therefore carries its own checker, witnesses, and consistency tests, and does not rely on the correctness of the emulator that produced it.

11 Discussion

11.1 A certification reading: scoped integrity evidence

Threat model. The protected asset is the *logical computational state* held in the memory (the k logical qubits' joint Pauli frame and any computation built on it). The trust boundary runs around the QEC pipeline — code instance, extraction schedule, decoder, and physical layout: everything inside is assumed to implement the pinned specification (verified by artifact digests against the trusted reference archive); everything outside — fabrication defects, calibration drift, control electronics, an adversary with physical access — is untrusted. The adversary of Definition 2 has full knowledge of code, schedule, and decoder, places at most W elementary faults non-adaptively, and cannot read syndrome outcomes before placement or influence native noise or decoding. Two scope rules are essential. (i) *QEC is not authentication*: an error-corrected memory does not bind identity, provide unforgeability, or detect all tampering — quantum authentication is a strictly stronger, cryptographically separate primitive [44, 45]; our bounds concern logical-error probability under $\mathfrak{A}_1^{\mathrm{st}}(W)$, nothing more. (ii) *Digest sealing is artifact integrity*: SHA-256 digests pin circuits, seeds, and results against the trusted reference archive; they are not device attestation and do not authenticate the hardware.

Scoped evidence profile. v23 organized the evidence as a five-level assurance ladder (QMAL-1–5); we replace it with an explicit *scoped evidence profile*: each artifact is valid only inside its

declared scope, and no hierarchical "level" is implied by accumulation. Each entry declares (*code*, *circuit/schedule*, *decoder*, *noise envelope*, *adversary*, *calibration interval*, *confidence*):

- *Code certificate* (O1–O5, Section 10): BB72 $[[72, 12, 6]]$ exact; algebraic matrices; no circuit, noise, or device scope.
- *Radius exhaustiveness* (O4): BP+OSD corrects all 2,629/2,629 weight-≤ 2 patterns per sector — ideal single-round phenomenological noise, no measurement faults. Under syndrome noise the v24 injection-boundary study (Table 3) covers a *frozen probe set*: 13 weight-≤ 2 data masks $\times$ 8,000 trials plus the 8 certified weight-3 halves $\times$ 4,000 trials, in a single-round setting with sampled noise — this is sampled probe evidence, not spacetime-fault or measurement-fault coverage, and the noiseless exhaustive test cannot establish the latter either.
- *Benchmark grid*: Tables 5–8, 11, 12; phenomenological noise, BP+OSD vs tested baselines, pinned seeds; fixed-sample-size grids use Wilson 95% intervals (exact Clopper–Pearson at zero counts), the adaptive ladder uses time-uniform confidence sequences; joint statements are Bonferroni-adjusted per displayed family only; the 10^6-trial rare-event refinement bounds BB288 LFR $< 2.996 \times 10^{-6}$ at $p \leq 0.004$.
- *Device injection campaigns*: REP3/5/7 on ibm_marrakesh (two calibration cycles, 2026-08-28 and 2026-09-06) and REP5 on ibm_fez (one cycle); adversary $\mathfrak{A}_1^{\mathrm{st}}(W)$, $W \leq t{+}1$; noise envelope = pinned calibration-snapshot model (λ_N readout-only Poisson); fixed-N arms, Wilson 95% per arm (chronology in Appendix B).
- *Active provenance* (Appendix A): calibration-informed placement on ibm_marrakesh *only* (CMH one-sided $p = 0.042$); v23's statement of a two-device demonstration overclaimed — the fez campaign replicated the injection bounds, not the placement effect.
- *Kunlun DEM re-analysis* (Section 8): model-level evidence only — Monte-Carlo sampling of the authors' released DEM, five decoders; no new hardware data.

Within this profile, syndrome-only monitoring is provably insufficient as an integrity certificate (Proposition 3): integrity evidence at the logical level requires injection probing of the kind reported in Appendices B–B.1. A live BB72 injection campaign awaits a publicly accessible BB-capable device.

The 2026 decoder landscape changes the baseline. The paired grid (Section 6) shows that BP+OSD's remaining competition is no longer the matching family but its own descendants: Relay-BP ties BP+OSD on identical instances while being designed for real-time hardware, and BP-augmented symmetry matching ties both. The circuit-level arms (Section 8) confirm on a released hardware DEM that graphified matching is 2.0–2.4$\times$ worse than BP+OSD where it matters (round 1), and Appendix A demonstrates on hardware that *where* a code line is placed is a measurable logical-rate lever ($p = 0.042$, CMH-pooled, four configurations).

Tested graphified baselines have substantially higher LFR than BP+OSD. Finite risk ratios of 67.8$\times$–595$\times$ for BB72 at 10,000 trials — refined at $p = 0.004$ by the 10^6-trial rare-event campaign to 371.9$\times$ [312.7, 442.3] (BB72), 17,500$\times$ [9,678, 31,643] (BB144), and $>$134,000$\times$ at joint $\geq 95\%$ confidence (BB288) — show that the tested DEM-chain MWPM and UF-BFS graphified baselines have substantially higher LFR than BP+OSD on the submitted BB instances. The emulator's `MWPMDecoder` provides a documented DEM-chain implementation for BB codes that handles the weight-3 column constraint, the odd-parity syndrome problem, and is fully transparent about its approximation. Any future study reporting an MWPM baseline for BB codes must specify the hyperedge treatment.

Hardware gap model-attributed. The ablation study (Section 9) provides simulation-model attribution of the CZ/MS performance gap. SWAP removal lowers CZ LFR from 0.0100 to 0.0050, while removing measurement noise lowers it from 0.0100 to 0.0023. These changes show that both routing and readout terms are material in the specified model. When MS device parameters are replaced by CZ-matched gate error, within this simulation model, the MS LFR rises from 0.00233 (MS full) to 0.00633 (MS with CZ-matched gate error), consistent with gate-decoherence parameters being a significant simulated contributor to the CZ/MS gap.

Graph-based decoders underperform BP+OSD on tested BB instances. The UF-BFS experiments (Section 5) show that both tested graphified baselines (DEM-chain MWPM and UF-BFS) underperform BP+OSD. UF-BFS uses a different greedy graph-decoding rule but the same weight-3-to-edge reduction, so the result is evidence against these two graphified baselines, not against all possible graph or matching-family decoders. The consistent ranking BP+OSD $\ll$ MWPM-DEM $<$ UF-BFS at all tested p values indicates that the submitted graphification loses information that BP+OSD uses natively; correlated or hypergraph-aware alternatives remain future work.

BP+OSD sub-threshold consistency. The repeated-cycle results across three code sizes are consistent with sub-threshold BP+OSD operation at $p \leq 0.008$. The emulator's space-time `run_repeated_cycle` runner (Eq. 1) enables this comparison in a single function call with full provenance.

Reproducibility as a scientific contribution. The QEC Emulator's provenance layer (Section 3, Eq. 2) provides the scripts, seeds, data files, and SHA-256 manifests required for independent reproduction in the declared software environment (ldpc 2.4.1, pymatching 2.4.0). The SHA-256 digests in Section 3.3 are live checks: any reader can instantiate `BB72()`, call `hx_sha256()`, and confirm they are computing with the same matrix.

12 Limitations

1. Pauli-weight-bounded threat class only; no cryptographic composability.
2. RC fixed grid at 2,000/500 trials, extended to 20,000-trial adaptive campaigns (Table 12); still not a finite-size scaling analysis.
3. MWPM-DEM is not an optimised MWPM baseline; hyperedge-aware alternatives (Relay-BP, belief-matching) are now included in Sections 6–8 and confirm the ordering, but an optimised correlated-matching implementation remains future work.
4. Hardware ablations (Section 9) use simulation only; the active-provenance A/B (Appendix A) is real-device execution but covers one device, one calibration cycle, and four repetition-code configurations, with a disclosed detection-rate dissociation on the REP5 arms.
5. The paired grid (Section 6) is single-round code-capacity noise on one code instance; the circuit-level corroboration (Section 8) covers one released hardware DEM and one synthetic BB72 circuit. Gate decoherence parameters from published characterisations may differ from actual on-device values.
6. The decoder-dependent ACE gap is bounded via the corrected Pauli-channel bridge (Proposition 1), which requires the either-sector endpoint and a Pauli channel; the coherent twirling residual is closed analytically for REP3 (Remark 9): $\theta_L = -2\theta^3 + O(\theta^5)$ exactly, the twirl gap is $D(E_L, \text{twirl}) = 2|g| \approx 4\theta^3$, and the diamond-norm bridge degrades as $2/(3\theta)$ outside the Pauli class. Extension of the analytic closure beyond REP3 is open.

7. The injection-budget device test (Appendix B) covers repetition memories; the consolidation (Appendix B.1) replicates it on two devices and two calibration cycles and extends the cliff to d=7, but larger distances and CSS-code injection remain open. The native fault-count distribution $\pi(k)$ entering Corollary 3 is calibration-estimated, not fitted to the results; stochastic arms cluster shots within 50 pinned patterns per cell; and the weight-$(t{+}1)$ arms saturate at 90–98% rather than 100% owing to native-noise rescue events. Corollary 2 covers general CSS memories under the spacetime fault model with exact minimum-weight decoding (Remark 6 delineates which conclusions implemented decoders inherit); a circuit-level spacetime-distance proof for BB memories with hook errors (beyond the certified single-round $t = 2$) is open. The repetition-code campaigns validate the bound structure and the calibration-to-model methodology; quantitative transfer of REP-calibrated noise models to BB-hardware performance predictions is not claimed and would require a separate BB-device modelling argument.

8. Routing bound uses LFR as proxy for diamond-norm (heuristic; the proxy is closed analytically for REP3 in Remark 9; uncalibrated for other codes).

9. The distance-ladder second rung (BB144→BB288) is statistically unresolvable at 20,000 trials per point under aligned priors (both codes failure-free at most grid points), so Λ is reported only for the first rung and the two-step combination, with one-sided bounds where denominators are zero.

10. The symatch comparison builds matching graphs with a fixed $p_{\text{phys}} = 0.001$ prior at all grid rates; the matched-prior control with an equal documented tuning budget is deferred to the confirmatory phase, and all matching conclusions are limited to the tested construction (Section 6).

11. The v23 circuit-level WS-C campaign used reversed CSS extraction conventions (C01); its data are invalidated and rerun with corrected circuits and independent deterministic verification (`wsc_verify_report.json`, four test groups). All v24 circuit-level numbers derive from the corrected circuits only.

13 Conclusion

We presented the ACE framework for qLDPC memories, formalized as an injection-budget game with an operational, device-measurable value — proved for repetition memories, extended to general CSS memories under the spacetime fault model (Corollary 2), and certified for BB72 at $t = 2$ — and supported by reproducible evidence layers produced by the open-source QEC Emulator v1.2.4 [1] together with two hardware campaigns on ibm_marrakesh. The central findings are: (i) the tested DEM-decomposed MWPM and UF-BFS graphified baselines have substantially higher LFR than BP+OSD on the submitted BB instances (finite RR 67.8× to 595× for BB72 at 10^4 trials, refined at $p = 0.004$ to 371.9×, 17,500×, and >134,000× at joint $\geq$ 95% confidence for BB288 by the 10^6-trial rare-event campaign); (ii) BP+OSD exhibits finite-size evidence of sub-threshold behavior at $p \leq 0.008$ under d-round repeated-cycle phenomenological noise across BB72/BB144/BB288; (iii) the CZ/MS hardware-noise-model gap is simulation-model attributed: SWAP routing, measurement noise, reset noise, and gate-decoherence parameters all materially affect the simulated gap (idle-noise removal leaves the CZ arm unchanged at 30/3,000, so no idle-noise contribution is claimed), with reset sensitivity quantified and shown to be logically distinct from readout noise; (iv) the BB72 certificate exhaustively establishes $d = 6$ exactly and $t = 2$; (v) on the pinned paired grid, bare symmetry matching is non-viable while its BP-augmented form, BP+OSD, and Relay-BP are equivalent within a ±0.5% paired margin (margin adopted retrospectively, not pre-registered; and equivalence at a margin is not point-null equality), and predecoding is redundant under BP guidance

on these instances; (vi) at circuit level the pairing separates — on the corrected extraction (C01 repair) Relay-BP fails zero of 20,000 shots at $p = 0.001$ (exact CP UB 1.50×10^{-4}) while BP+OSD fails 51 (LER 2.55×10^{-3}, Wilson CI $[1.94, 3.35] \times 10^{-3}$), despite the code-capacity tie — and the tested Stim-based matching toolchain cannot be instantiated natively on this weight-6-check circuit (coverage failure 69.1%); both statements are scoped to the tested reductions and configurations; (vii) on the released Kunlun hardware DEM the matching-family penalty is 2.0–2.4× at round 1 and decays with depth, and Relay-BP matches 10^4-iteration BP+OSD at $\sim 35\times$ lower wall time; (viii) a calibration-informed placement choice lowered the logical error rate in four of four repetition-code configurations on ibm_marrakesh (CMH pooled $p = 0.042$); and (ix) the theory layer was taken to device under a device injection test with pinned protocols and pinned masks: weight-t worst-case injections remained within the composed correctability bound and weight-$(t{+}1)$ injections collapsed the memory at 92–96%; all stochastic cells are consistent with the mask-conditioned thresholds computed from the pinned masks (the conditioning estimand was corrected retrospectively after inspection of the mask issue — the chronology is disclosed in Appendix B — and the v23 falsification verdict against the population-averaged independence form is withdrawn under the corrected estimand; prospective confirmation of the composed model on fresh data remains open) — a pattern that replicated on a second calibration cycle and a second device (ibm_fez) and extends to distance 7, where the weight-4 cliff remains sharp (96%). Three additions sharpen the framework: (x) syndrome monitoring is a strictly weaker integrity certificate than logical-level injection probing (Proposition 3), with the anti-correlated REP5 A/B pair as the measured instance; (xi) injection probing over an enumerated attack set gives exact certified values where the enumeration is complete ($t = 2$ radius; the noiseless weight-≤ 2 decoder-radius test) and honest intervals $[L_W, U_W]$ with $U_W = 1$ where it is not (Remark 2, Table 2), and the adversarial work factor against the BB ladder reaches weight 3–9 versus 1 for the rate-matched independent-patch baseline (Corollary 4); (xii) the $d = 6{\to}12{\to}18$ ladder suppresses the per-block logical rate with observed cube-root-per-$\Delta d{=}2$ factors $\Lambda = 3.9$–5.4 at $p \geq 0.006$, with confidence-sequence lower bounds $\Lambda > 1.2$–2.2 valid under the adaptive stopping rule (Section 7; phenomenological model, block $= d$ rounds — not a per-cycle factor and not a matched protocol to the circuit-level hardware value $\Lambda = 2.14$ of [31]); and the evidence assembles into the scoped integrity-evidence profile of Section 11.1. All data, code, seeds, and SHA-256 fingerprints are packaged for Zenodo archival; the archive is published at doi:10.5281/zenodo.20588043. The QEC Emulator is designed for extension: new code instances, decoder backends, noise models, and benchmark protocols require only interface implementation without modifying the core package.

Claim-to-evidence map

Claim	Type	File / tool	Limitation
Central estimand $\lambda_{\mathcal{M}}(W)$ (fixed-implementation adversarial risk)	Definition	Def. 4, Rem. 1	Worst case over fault sets; no defender optimization
Minimax saddle point	Theorem	–	Background: compact CPTP strategy sets; existence only
Fidelity floor	Theorem	–	Background: ideal recovery; bounded-weight Pauli class
Independence-form LB	Proposition	`data/minimax_lower_bounds.csv`	Population-average form; device falsification *withdrawn* under mask-conditioned estimand (v24)
Injection-budget game	Definition	`theory_test/`	Operational value; non-adaptive adversary
Four-category decoder endpoint	Definition	Def. 3	Repairs v23 conflation of invalid corrections with logical failures

continued on next page

Claim	Type	File / tool	Limitation
Spacetime bound	Theorem	`theory_test/`	Protected sector (bit-flip + meas. faults); repetition memory proved; BB72 certified at t=2 only
CSS spacetime extension	Corollary	(proof in text)	Spacetime fault model; exact min-weight decoding; no hook errors
Diamond bridge (corrected)	Proposition	`data/twirling_residual_v24.csv`	Either-sector endpoint; Pauli approximation; residual closed analytically (Rem. 9)
Device-aware stoch. bound	Corollary	`theory_test/results/ theorytest_results.csv`	$\pi(k)$ calibration-model prediction (readout-only Poisson), not a conservative envelope
Scaling lemma	Lemma	–	Independent-patch counting comparison; matched-rate for BB72 only
Routing bound	Numeric example	–	Heuristic numeric example; LFR is not a diamond-norm bound
Emulator architecture	Software	`qec_emulator` package	Software infrastructure, not a theorem
BB72 $d = 6$ exact	Exhaustive	`data/bb72_certificate_ report.json`	O3b excludes $d \leq 5$; O5 exhibits $84 + 84$ weight-6 logicals
BB72 $t = 2$	Exhaustive	`verify_certificate.py`	Radius-2 correctability only
Syndrome-oracle insufficiency	Proposition	Appendices A–B	Decision-level separation with explicit budget $W \geq \lceil w(L)/2 \rceil$; not a hardness claim
Injection certification	Remark (was Cor.)	`theory_test/`, `theory_test_ consol/`	Three honest outputs: enumerated-set certificate / model-conditional bound / challenge test with $U_W = 1$
Adversarial work factor	Corollary	(proof in text)	Deterministic tie-breaking + radius-complete decoder assumed; BB144/BB288 at their own lower rates
Injection boundary, 4 decoders + MW-lookup	Paired MC	`data/injection_boundary_ decoders_v24.csv`, `inj_radius_ noiseless_v24.csv`, `injection_ boundary_mcnemar_v24.csv`	v24 frozen paired protocol; supersedes v23 file
Twirling residual (REP3)	Analytic	`data/twirling_residual_v24.csv`	Closed form $2\|g\|$, $2\sqrt{p_L^2 + g^2}$; archived variational values coincide to rounding
Distance-ladder Λ	Empirical	`data/lambda_ladder_v24.csv`	Cube root of *block*-failure ratio; phenomenological; second rung unresolvable at 20k trials
SR BP+OSD (3 codes)	Monte Carlo	`data/single_round_all_ codes.csv`	Single-round model
SR MWPM-DEM (3 codes)	Monte Carlo	`data/single_round_all_ codes.csv`	Tested DEM-chain baseline only
SR UF-BFS (2 codes)	Monte Carlo	`data/single_round_uf_ decoder.csv`	Second graphified baseline; not hyperedge-aware
RC BP+OSD (3 codes)	Monte Carlo	`data/repeated_cycle_*_v24.csv`	v24 aligned priors; 500–2,000 trials; phenomenological model
RC adaptive extension	Monte Carlo	`data/repeated_cycle_adaptive_ *_v24.csv`	v24 aligned priors; 4,000–20,000 trials; pinned stopping
Exact REP stratification	Exact	`data/rep_exact_masks_v24.csv`, `rep_exact_lfr_v24.csv`	All 168 masks enumerated; zero sampling variance; analytic == tableau
Rare-event BP+OSD 10^6	Monte Carlo	`data/rare_event_bposd.csv`	Single-round $p \in \{0.002, 0.004\}$ only; exact CP zero-failure bounds
CZ/MS hardware gap	Simulation	`data/hardware_ablations.csv`; `data/hardware_three_condition_ comparison.csv`	Noise-model sensitivity; no real-device causality; no idle-noise contribution measurable
Reset sensitivity	Monte Carlo	`reset_sensitivity.csv`	Single-cycle circuit model; 3,000 trials
Paired decoder grid	Paired MC	`comparator/results/comparator_ grid_final.csv`	Code-capacity, BB72 only; symatch fixed $p_{\text{phys}} = 0.001$ prior disclosed

continued on next page

Claim	Type	File / tool	Limitation
Symatch weight-1 diagnostic	Exact	`comparator/results/symatch_w1_diagnostic_v24.json`	Corrects v23 counts to 24/72 (X), 15/72 (Z) on this instance
Kunlun DEM re-analysis	Reuse	`comparator/results/kunlun_cmp.csv`	Released-DEM Monte-Carlo re-analysis; not raw shots
WS-C circuit point	Monte Carlo	`comparator/results/wsc_circuit_v24.csv`, `wsc_verify_report.json`	v24 corrected CSS extraction (C01), independently verified; one synthetic circuit
Active provenance A/B	Hardware	`active_provenance/`	Exploratory pilot: CMH one-sided p=0.042, two-sided 0.084; one device (marrakesh), one calibration cycle
Injection-budget device test	Hardware	`theory_test/`	Repetition memories; mask-conditioned estimand (v24 reanalysis)
Consolidation (2nd cycle, 2nd device, d=7)	Hardware	`theory_test_consol/`	Repetition memories; replication; stochastic cells consistent under conditioned thresholds

Data and code

Archive structure. This paper uses three distinct software artifacts: (1) *Lightweight reproduction archive* — the primary data files, runner scripts, verification scripts, environment files, Dockerfile, and SHA-256 manifest; it is self-contained and is reproduced by running `python reproduce_all.py` in the declared environment. (2) *QEC Emulator v1.2.4* — the reusable Python package (`pip install qec-emulator`); its five layers correspond to `codes.py`, `decoders.py`, `noise.py`, `benchmark.py`, and `provenance.py` as described in Section 3. (3) *Legacy supplementary data* — thirteen earlier-experiment CSVs in `data/legacy/`; these are not cited in manuscript tables but are preserved for completeness.

Archive: prepared for Zenodo deposition; doi:10.5281/zenodo.20588043. QEC Emulator v1.2.4 [1]: repository link to be inserted after public release. Primary data files:

- `single_round_all_codes.csv` (42 rows); `repeated_cycle_all_codes.csv` (21 rows)
- `rare_event_bposd.csv` (6 rows, 10^6 trials each)
- `repeated_cycle_adaptive_*.csv` (adaptive RC extension, 4,000–20,000 trials per row, including the `..._p010_*` point)
- `sha256_manifest.json` (manifest-tracked file count reported by the archive verifier); `README_REPRODUCE.md`
- `hardware_ablations.csv` (8-condition ablation); `single_round_uf_decoder.csv` (UF-BFS 14 points)

Elevation-layer artifacts (this version): `comparator/` (paired grid, McNemar tests, Kunlun and WS-C result CSVs, failbit arrays, timing probe), `active_provenance/` (A/B counts, circuits, backend snapshot, job records, SHA-256 manifest), `theory_test/` (hash-pinned predictions, 312-arm circuit digests, raw counts, verdict record for the injection-budget device test), `theory_test_consol/` (the 2026-09-06 consolidation: pinned predictions and estimator, 324-arm circuit digests, raw counts, verdict record, both device snapshots), and `reset_sensitivity.csv` (P1-11 closure run).

Author contributions

S.H. AlDaajeh conceived the ACE framework, proved all theorems, designed and implemented QEC Emulator v1.2.4, ran all experiments, and drafted the manuscript. [AUTHOR 2–5: contribution statements to be completed, e.g. per the CRediT taxonomy.] All authors reviewed and approved the manuscript.

Conflict of interest

The authors declare no competing interests.

A Active provenance: a controlled defect-placement experiment

Calibration data are typically consumed passively; here the measured defect landscape is used *actively*, as an independent variable. From a full calibration snapshot of `ibm_marrakesh` (156 qubits, 2026-08-28), we selected two physical 11-qubit lines whose measured defect profiles are matched on every coordinate except one deliberately varied difference (the B line avoids a qubit whose readout error is 3.43× the line median). Selection was scripted and digest-pinned before any QPU job was submitted (selection digest `b2c7c36e...f2aa87d`), following the defect-aware-operation paradigm of [33], whose 2.8× exclusion-side suppression motivates treating placement itself as an experimental control.

Eight jobs (REP3/REP5 repetition codes × X/Z bases × A/B lines), 10,000 shots each, were executed inside a single 40-minute window (5 s QPU charge time per job; all QPY circuits, job IDs, backend snapshot, and raw counts archived and hashed in `MANIFEST.json`). Table 19 reports the logical-rate counts; Figure 5 visualises the two physical lines, the logical error rates with the CMH pool, and the per-detector event rates that carry the REP5 dissociation.

Table 19: Active-provenance A/B on `ibm_marrakesh`, 10,000 shots per arm, Wilson 95% CIs. B (defect-avoiding) is lower in all four configurations; CMH pooled $z = -1.73$, one-sided $p = 0.042$, common odds ratio 1.43.

Config	A-line LER	B-line LER	z_{B-A}
REP3-Z	$19/10^4 = 1.90 \times 10^{-3}$ $[1.22, 2.97] \times 10^{-3}$	$15/10^4 = 1.50 \times 10^{-3}$ $[0.91, 2.47] \times 10^{-3}$	-0.69
REP3-X	$30/10^4 = 3.00 \times 10^{-3}$ $[2.10, 4.28] \times 10^{-3}$	$21/10^4 = 2.10 \times 10^{-3}$ $[1.37, 3.21] \times 10^{-3}$	-1.26
REP5-Z	$1/10^4 = 1.0 \times 10^{-4}$ $[0.18, 5.66] \times 10^{-4}$	$0/10^4$ $[0, 3.84] \times 10^{-4}$	-1.00
REP5-X	$7/10^4 = 7.0 \times 10^{-4}$ $[3.39, 14.4] \times 10^{-4}$	$4/10^4 = 4.0 \times 10^{-4}$ $[1.56, 10.3] \times 10^{-4}$	-0.91

Limitation (disclosed). On REP5 the B line runs *hotter* at the syndrome level (per-detector event rates higher, $z = +3.69/+3.87$) yet *colder* logically — a detection-rate/logical-rate dissociation attributable to an under-modeled readout on one B-line qubit. The REP5 arms are therefore conservative for the placement hypothesis; all four arms are reported without selection.

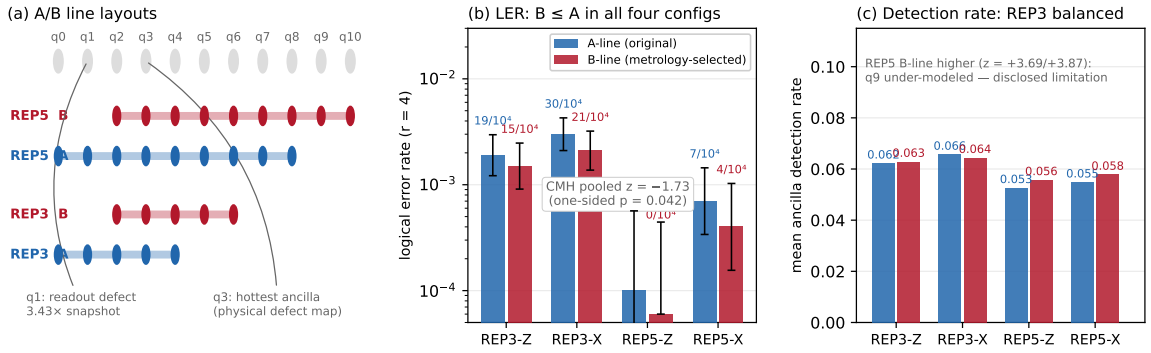


Figure 5: Active provenance: (a) A/B line layout with defect annotations; (b) logical error rates with Wilson CIs and the CMH pool; (c) per-detector event rates with the REP5 dissociation disclosed.

B Pre-registered device test of the injection-budget bound

We test Theorem 3 and Corollary 3 on ibm_marrakesh (156-qubit Heron r2) with REP5 (d=5, t=2) and REP3 (d=3, t=1) memories, both sectors, R=4 rounds. The adversary injects Pauli faults at round 2 (Z-sector: physical X gates; X-sector: basis-conjugated physical X gates, H–X–H, realizing Z-type faults on the $|+_L\rangle$ memory; each injected fault is thus implemented with native gates in both sectors). Layouts are the metrology-selected B-lines of Appendix A, fixed across all arms so that injection is the only varying factor. Predictions, decision rules, seeds, layouts, and circuit digests were pinned (SHA-256 `759169c2...c74653cd5`) *before* submission; 312 arms (180,000 shots) ran in four Sampler jobs (`da8um21qtnsc73d0v3ug`, `da8um2ce74ec73aj4n3g`, `da8um31qtnsc73d0v400`, `da8um3ke74ec73aj4n50`); the full noiseless pipeline was validated end-to-end (22/22 arms) before submission. $\pi(k)$ entering the composed bounds is estimated from the calibration-snapshot fault model (λ_N=0.47 faults/shot REP5, 0.25 REP3): $\pi(1..3) = 0.375, 0.081, 0.012$ (REP5) and $0.221, 0.027, 0.002$ (REP3). **Model status of π:** these tails are *predictions of a pinned model* — a Poisson fault-count law whose intensity λ_N sums readout-error locations only (gate/idle terms enter at second order on Heron r2). They are not conservative envelopes over calibration drift; their empirical content is tested by the arms below, not assumed. Scope: this campaign tests the injection-budget bound and the calibration-to-model workflow on repetition memories; it is not a measurement of BB-code hardware performance, and no quantitative REP→BB transfer is claimed.

Table 20: Deterministic worst-case injection arms (10,000 shots each). "Bound": Theorem 3(c) with calibration π.

arm	sector	failures	LER	bound	verdict
REP5 base	Z	0	0.0000	0.012	holds
REP5 base	X	2	0.0002	0.012	holds
REP5 w=2 worst	Z	414	0.0414	0.375	holds
REP5 w=2 worst	X	506	0.0506	0.375	holds
REP5 w=3 worst	Z	9590	0.9590	—	sharp
REP5 w=3 worst	X	9598	0.9598	—	sharp
REP3 base	Z	14	0.0014	0.027	holds
REP3 base	X	28	0.0028	0.027	holds
REP3 w=1 worst	Z	276	0.0276	0.221	holds
REP3 w=1 worst	X	318	0.0318	0.221	holds
REP3 w=2 worst	Z	9292	0.9292	—	sharp
REP3 w=2 worst	X	9240	0.9240	—	sharp

Table 21: Stochastic injection arms, REP5 (50 pinned variants × 200 shots per cell). "Naive (pop. avg.)": v23's independence-form threshold $\delta_0 + \Pr[\text{Bin} \geq 3] + \text{tol}$. "Naive (mask-cond.)": the mask-conditioned estimand, conditioning on the realized pinned masks, $\delta_0 + \frac{1}{50}\sum_v \mathbf{1}[w(\text{mask}_v) \geq t{+}1] + \text{tol}$ (`stoch_reanalysis_v24.csv`); the verdict is read against this column. The conditioning estimand was adopted retrospectively, at reanalysis after inspection of the realized masks (Table 23). Composed: Corollary 3 with calibration π.

p	sector	failures	LER	naive (pop. avg.)	naive (mask-cond.)	composed	verdict
0.01	Z	1	0.0001	0.0024	0.0024	0.016	holds
0.01	X	2	0.0002	0.0027	0.0027	0.016	holds
0.05	Z	6	0.0006	0.0035	0.0024	0.035	holds
0.05	X	9	0.0009	0.0039	0.0027	0.035	holds
0.10	Z	211	0.0211	0.0109	0.0224	0.070	holds
0.10	X	388	0.0388	0.0113	0.0427	0.070	holds

Three findings. (i) *The distance cliff is real and sharp:* weight-t worst-case injections stay within the composed correctability bound on both distances and both sectors, while weight-$(t{+}1)$ injections collapse the memory at 92.4–96.0% logical failure; the shortfall from 100% is native-noise rescue of order $\pi(1)$. (ii) *Injection amplifies native noise exactly as the chain argument predicts:* the REP5 w=2 arms sit 100–250× above baseline but an order of magnitude inside the composed bound — the first quantitative measurement of adversary–noise composition on a quantum memory. (iii) *Stochastic arms and the correct estimand.* v23 reported the p=0.10 cells as falsifying the independence-form stochastic bound (Proposition 2) at 2.1× (Z) and 3.8× (X). That verdict compared realized LER against the *population-averaged* threshold; the pinned protocol, however, fixes 50 masks, so the correct threshold for the pinned ensemble is *mask-conditioned* (an estimand correction adopted retrospectively, after inspection — Table 23): the X ensemble realizes two weight-3 masks versus 0.43 expected under $\text{Bin}(5, 0.1)$ (realized tail 0.04), the Z ensemble one (tail 0.02). Against the conditioned thresholds (0.0224 Z, 0.0427 X) the realized LERs (0.0211, 0.0388) do *not* exceed, and the same holds for all six cells in both campaigns (`stoch_reanalysis_v24.csv`, 12 cells, margins 0.0013–0.0039): **the falsification verdict is withdrawn**. What survives is a genuine observation — the population-average form undershoots realized-mask risk whenever the realized ensemble is tail-heavy — which *motivated* the composed Binomial⊗Poisson model (Corollary 3) as a discovery. That model is therefore reported as discovery, not as a pre-registered prediction; its confirmatory test is the consolidation campaign of Appendix B.1. The device-aware composed form holds on all six stochastic arms

and all eight bounded deterministic arms under either estimand.

Security corollary. Combined with the REP5 detection-rate dissociation of Appendix A (placement moves the detection rate, not the logical rate), these results imply that syndrome-level statistics cannot certify memory integrity: a memory is certifiably (W, η)-integrity-secure — $\lambda_{\mathcal{M}}(W) \leq \eta$ under $\mathfrak{A}_1^{\mathrm{st}}(W)$ — only via logical-level injection probing of the kind reported here.

B.1 Replication: second calibration cycle, second device, distance 7

A pre-registered consolidation campaign (predictions, decision rules, seeds, layouts, estimator, and digests pinned before submission, SHA-256 `cef236ce...de53aa1`) re-ran the test on ibm_marrakesh on a second calibration cycle (2026-09-06; 240,000 shots in four Sampler jobs `daelek5nj4cs73afaqpg`, `daelekd1ierc738mai50`, `daelel642tqs73auss4g`, `daeleljdd5gc73d8pcrg`), extended it to REP7 (d=7, t=3; worst-case central strings w=3 and w=4), and replicated the REP5 deterministic arms on a second device, ibm_fez (60,000 shots; jobs `daelelt1ierc738mai7g`, `daelembdd5gc73d8pct0`). Layouts were re-selected from the pinned 2026-09-06 snapshots by the metrology rule of Appendix A (the REP5 rule returned the campaign-1 line); the stochastic masks are bit-identical to campaign 1. The native-fault estimator is pinned explicitly, $\lambda_N = \sum_{\mathrm{data}} \epsilon_{\mathrm{ro}} + R \sum_{\mathrm{anc}} \epsilon_{\mathrm{ro}}$ with symmetric readout ϵ_{ro} (readout locations dominate the Heron-r2 spacetime budget; gate/idle terms are second order): λ_N=0.088, 0.222, 0.347 (REP3/5/7, marrakesh) and 0.453 (REP5, fez) — the 2026-09-06 marrakesh calibration is substantially cleaner than that of 2026-08-28 (λ_N=0.25, 0.47). Noiseless end-to-end validation passed on all 42 checked arms before submission.

Table 22: Consolidation deterministic arms (10,000 shots each): REP7 extension (ibm_marrakesh) and second-device REP5 replication (ibm_fez). "Bound": Theorem 3(c) with the pinned calibration π.

device	arm	sector	failures	LER	bound	verdict
marrakesh	REP7 base	Z	0	0.0000	0.0005	holds
	REP7 w=3 worst	Z	568	0.0568	0.293	holds
	REP7 w=4 worst	Z	9691	0.9691	—	sharp
	REP7 base	X	1	0.0001	0.0005	holds
	REP7 w=3 worst	X	675	0.0675	0.293	holds
	REP7 w=4 worst	X	9599	0.9599	—	sharp
fez	REP5 base	Z	1	0.0001	0.011	holds
	REP5 w=2 worst	Z	484	0.0484	0.364	holds
	REP5 w=3 worst	Z	9768	0.9768	—	sharp
	REP5 base	X	6	0.0006	0.011	holds
	REP5 w=2 worst	X	683	0.0683	0.364	holds
	REP5 w=3 worst	X	9745	0.9745	—	sharp

Four findings. (i) *Full replication.* On the new calibration cycle every campaign-1 verdict reproduces: all bounded deterministic arms sit 4–8× inside the composed bound on both distances and sectors (REP3 w=1: 0.0102/0.0202 vs bound 0.084; REP5 w=2: 0.0333/0.0436 vs 0.199), all weight-(t+1) arms remain sharp (0.898–0.974), and all six stochastic cells track campaign 1 within statistical noise (at p=0.10: 0.0202 vs 0.0211 Z; 0.0403 vs 0.0388 X). The campaign-1 excess over the population-averaged independence-form threshold replicates (1.9× Z, 3.6× X), and, as in campaign 1, no cell exceeds the mask-conditioned threshold (0.0224 Z, 0.0426 X) — for this consolidation the estimand was part of the pre-submission pinning, so no falsification occurs in either campaign under the correct estimand (`stoch_reanalysis_v24.csv`). (ii) *The*

cliff has a third distance point. REP7 behaves as Theorem 3 demands: weight-3 worst-case injections stay inside the bound (0.057/0.068 vs 0.293) while weight-4 collapses the memory at 96.0–96.9%; the cliff is sharp at d=3, 5, 7. (iii) *Cross-device.* The REP5 pattern reproduces on ibm_fez with its own pinned calibration (λ_N=0.453): bounds hold at 5–7× margin, sharpness at 97.5–97.7%. (iv) *Mask-level composition.* With the cleaner calibration the population-average binomial-form composed bound tightens (0.0335 for the X-sector p=0.10 cell), and the realized LER (0.0403) exceeds that population-average form by 1.20×; the excess is entirely accounted for by the realized pinned-mask ensemble — the X masks contain two weight-3 variants versus 0.43 expected under Bin(5, 0.1), contributing 0.0375 — and conditioning the composed bound on the realized masks (bound 0.0665) restores it: the composition law holds mask-by-mask, not only in binomial expectation. This is the same estimand correction as in campaign 1, applied to the composed model. Absolute LERs of bounded arms drift down coherently with the cleaner calibration (λ_N: 0.47→0.22), while cliff arms drift slightly up (less native-noise rescue of order $\pi(1)$), exactly as the model predicts. The consolidation artifacts (`theory_test_consol/`) are archived with the same digest discipline as campaign 1.

C Chronology and protocol-pinning ledger

Table 23 records, for every campaign in this work, what was frozen before data-taking and what was amended after inspection. Terminology: *pinned* means the artifact (protocol, masks, seeds, predictions, decision rules, digests) was hash-frozen before the corresponding runs; *retrospective* means the estimand or rule was adopted after inspecting data. Nothing in this manuscript is described as pre-registered beyond the pinnings listed here, and every estimand adopted after data inspection is labeled retrospective both here and at its point of use.

Table 23: Chronology ledger: date, campaign, what was frozen before data-taking, estimand status, result, and amendments. Hashes are SHA-256 prefixes of the pinned prediction bundles.

date	campaign	frozen before data-taking	estimand status	result / amendments
2026-08-28	placement A/B pilot (App. A)	line-selection rule and A/B pairing from the 2026-08-28 snapshot (`MANIFEST.json` 08:28 UTC) before the 40-minute run window	paired-LFR association (CMH), exploratory	4/4 configurations favor B-lines; pooled p=0.042, exact stratified p=0.0517; reported as descriptive pilot, no causal claim
2026-08-28	device campaign 1 (App. B)	predictions, decision rules, seeds, layouts, circuit digests hash-pinned (`759169c2...`); 22/22 noiseless validation passed pre-submission	pinned estimand: population-averaged thresholds and deterministic bound verdicts	deterministic bounds hold; weight-(t+1) sharp; stochastic p=0.10 cells exceed the pinned threshold 2.1–3.8× (verdict later withdrawn — next row)
2026-09	campaign-1 reanalysis (amendment)	— (analysis-only; no new device data)	mask-conditioned estimand adopted *retrospectively*, after inspection of the realized masks	falsification verdict withdrawn; composed Binomial⊗Poisson model developed as *discovery*; not part of the pre-submission pinning
2026-09-06	consolidation (App. B.1)	predictions, rules, seeds, layouts, estimator, digests hash-pinned (`cef236ce...`); stochastic masks bit-identical to campaign 1	mask-conditioned estimand *included* in the pre-submission pinning (carried over from the campaign-1 correction)	full replication on a second calibration cycle; REP7 (d=7) extension; second-device (ibm_fez) replication
v23 cycle	comparator grid; repeated-cycle fixed ladder	paired streams and grid frozen per archived run manifests	exploratory; ±0.5% equivalence margin adopted retrospectively for the archived grid (pinned only for the confirmatory protocol)	archived as exploratory; equivalence statements are at-margin, not point-null
v24 freeze	injection-boundary decoders; adaptive RC extension	frozen probe set and frozen native-error replay (`ibd_v24_freeze.json`); adaptive stopping rule pinned before launch	four-category endpoint; adaptive arms analysed with time-uniform confidence sequences	second-round-review repairs (2026-09): worst-case interval rule corrected (Remark 2); adaptive intervals regenerated with Robbins mixture sequences (`stats_intervals.py`); endpoint classifier made true-residual-based with a regression test (App. B endpoint audit)
2026-09	WS-C circuit arm (Sec. 8)	corrected extraction circuit (C01 repair) with invariance checks; six-rate grid pinned per-point with checkpointing	fixed-N arms, Wilson/CP per arm	single corrected point archived; the full corrected grid is regenerated from the pinned per-point runners and merged into Table 14

Standing consequence. Any future amendment enters this ledger with its date and pinning status; claims whose estimands are retrospective (the placement pilot, the campaign-1 mask conditioning, the archived-grid equivalence margin) are excluded from any confirmatory reading of this work.

D Illustrative routing arithmetic

Heuristic only; LFR underestimates diamond-norm error. $J = 8$, $b = 1$, $\Delta = 0.05$: $p_{\text{hit}} \leq 0.175$. CZ at $p = 5 \times 10^{-4}$: $(0.825)(0.905) = 0.747$. MS: $(0.825)(0.955) = 0.788$.

References

[1] Saleh H. AlDaajeh. QEC Emulator: Reproducible qLDPC benchmark suite with hash-anchored provenance, 2026. URL https://doi.org/10.5281/zenodo.20588043.

[2] Peter W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26(5):1484–1509, 1997. doi: 10.1137/S0097539795293172.

[3] National Institute of Standards and Technology. FIPS 203: Module-lattice-based key-encapsulation mechanism standard. Federal Information Processing Standards Publication 203, National Institute of Standards and Technology, 2024. URL https://csrc.nist.gov/pubs/fips/203/final.

[4] National Institute of Standards and Technology. FIPS 204: Module-lattice-based digital signature standard. Federal Information Processing Standards Publication 204, National Institute of Standards and Technology, 2024. URL https://csrc.nist.gov/pubs/fips/204/final.

[5] National Institute of Standards and Technology. FIPS 205: Stateless hash-based digital signature standard. Federal Information Processing Standards Publication 205, National Institute of Standards and Technology, 2024. URL https://csrc.nist.gov/pubs/fips/205/final.

[6] Sergey Bravyi, Andrew W. Cross, Jay M. Gambetta, Dmitri Maslov, Patrick Rall, and Theodore J. Yoder. High-threshold and low-overhead fault-tolerant quantum memory. *Nature*, 627:778–782, 2024. doi: 10.1038/s41586-024-07107-7.

[7] Swayangprabha Shaw and Narayanan Rengaswamy. Networked realization of quantum LDPC codes. *arXiv preprint arXiv:2604.25026*, 2026.

[8] Joschka Roffe, David R. White, Simon Burton, and Earl T. Campbell. Decoding across the quantum low-density parity-check code landscape. *Physical Review Research*, 2:043423, 2020. doi: 10.1103/PhysRevResearch.2.043423.

[9] Joschka Roffe. LDPC: a Python package for low-density parity-check codes. https://roffe.eu/software/ldpc, 2022.

[10] Oscar Higgott and Craig Gidney. Sparse blossom: correcting a million errors per core second with minimum-weight matching. *Quantum*, 9:1600, 2025. doi: 10.22331/q-2025-01-20-1600.

[11] Oscar Higgott and Craig Gidney. PyMatching v2: sparse blossom back-end. https://pymatching.readthedocs.io, 2025.

[12] Craig Gidney. Stim: a fast stabilizer circuit simulator. *Quantum*, 5:497, 2021. doi: 10.22331/q-2021-07-06-497.

[13] Jean-Pierre Tillich and Gilles Zémor. Quantum LDPC codes with positive rate and minimum distance proportional to the square root of the blocklength. *IEEE Transactions on Information Theory*, 60(2):1193–1202, 2014. doi: 10.1109/TIT.2013.2292061.

[14] Pavel Panteleev and Gleb Kalachev. Asymptotically good quantum and locally testable classical LDPC codes. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, STOC 2022, pages 375–388. Association for Computing Machinery, 2022. doi: 10.1145/3519935.3520017.

[15] Anthony Leverrier and Gilles Zémor. Quantum Tanner codes. *arXiv preprint arXiv:2202.13641*, 2022.

[16] Daniel Gottesman. Fault-tolerant quantum computation with constant overhead. *Quantum Information and Computation*, 14(15–16):1338–1372, 2014.

[17] Omar Fawzi, Antoine Grospellier, and Anthony Leverrier. Constant overhead quantum fault-tolerance with quantum expander codes. In *Proceedings of the 2018 IEEE 59th Annual Symposium on Foundations of Computer Science*, FOCS 2018, pages 743–754. IEEE, 2018. doi: 10.1109/FOCS.2018.00076.

[18] Matthias Christandl, Omar Fawzi, and Ashutosh Goswami. Fault-tolerant quantum computation with constant overhead for general noise. *PRX Quantum*, 6:040334, 2025. doi: 10.1103/k4cm-pp9p.

[19] Charles H. Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. In *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, pages 175–179, Bangalore, India, 1984.

[20] Peter W. Shor and John Preskill. Simple proof of security of the BB84 quantum key distribution protocol. *Physical Review Letters*, 85:441–444, 2000. doi: 10.1103/PhysRevLett.85.441.

[21] Christopher Portmann and Renato Renner. Security in quantum cryptography. *Reviews of Modern Physics*, 94:025008, 2022. doi: 10.1103/RevModPhys.94.025008.

[22] Robert Rand, Jennifer Paykin, and Steve Zdancewic. QWIRE practice: Formal verification of quantum circuits in Coq. In *Proceedings of the 14th International Conference on Quantum Physics and Logic*, volume 266 of *Electronic Proceedings in Theoretical Computer Science*, pages 119–132, 2018. doi: 10.4204/EPTCS.266.8.

[23] Kesha Hietala, Robert Rand, Shih-Han Hung, Xiaodi Wu, and Michael Hicks. A verified optimizer for quantum circuits. *Proceedings of the ACM on Programming Languages*, 5 (POPL):1–29, 2021. doi: 10.1145/3434318.

[24] Anthony Bordg, Hannah Lachnitt, and Yijun He. Certified quantum computation in Isabelle/HOL. *Journal of Automated Reasoning*, 65:691–709, 2021. doi: 10.1007/s10817-020-09584-7.

[25] Marco Lewis, Sadegh Soudjani, and Paolo Zuliani. Formal verification of quantum programs: Theory, tools, and challenges. *ACM Transactions on Quantum Computing*, 5(1): 1–35, 2023. doi: 10.1145/3624483.

[26] Tristan Müller et al. Improved belief propagation is sufficient for real-time decoding of quantum memory. *arXiv preprint arXiv:2506.01779*, 2025.

[27] Thilo Maurer et al. Real-time decoding of the gross code memory with FPGAs. *arXiv preprint arXiv:2510.21600*, 2026.

[28] Kaavya Sahay, Dominic J. Williamson, and Benjamin J. Brown. A matching decoder for bivariate bicycle codes. *arXiv preprint arXiv:2602.22770*, 2026. v2, June 2026.

[29] Oscar Higgott, Thomas C. Bohdanowicz, Aleksander Kubica, Steven T. Flammia, and Earl T. Campbell. Improved decoding of circuit noise and fragile boundaries of tailored surface codes. *Physical Review X*, 13:031007, 2023.

[30] Michele Pacenti, Nithin Raveendran, and Bane Vasic. Trapping sets of detector error models. *arXiv preprint arXiv:2608.11516*, 2026.

[31] Google Quantum AI and collaborators, Rajeev Acharya, et al. Quantum error correction below the surface code threshold. *Nature*, 638:920–926, 2025.

[32] Ke Wang, Zhide Lu, Chuanyu Zhang, et al. Demonstration of low-overhead quantum error correction codes. *Nature Physics*, 22:308, 2026. Dataset: Zenodo, CC-BY-4.0.

[33] Drouet et al. Error correction on an array of superconducting qubits with defective components. *arXiv preprint arXiv:2607.12118*, 2026.

[34] Michael E. Beverland, Malcolm Carroll, Andrew W. Cross, and Theodore J. Yoder. Fail fast: techniques to probe rare events in quantum error correction. *arXiv preprint arXiv:2511.15177*, 2025.

[35] Robert L. Kosut, Alireza Shabani, and Daniel A. Lidar. Robust quantum error correction via convex optimization. *Physical Review Letters*, 100:020502, 2008. doi: 10.1103/PhysRevLett.100.020502.

[36] Eric Huang, Andrew C. Doherty, and Steven T. Flammia. Performance of quantum error correction with coherent errors. *Physical Review A*, 99:022313, 2019. doi: 10.1103/PhysRevA.99.022313.

[37] Maurice Sion. On general minimax theorems. *Pacific Journal of Mathematics*, 8(1):171–176, 1958. doi: 10.2140/pjm.1958.8.171.

[38] Emanuel Knill and Raymond Laflamme. Theory of quantum error-correcting codes. *Physical Review A*, 55:900–911, 1997. doi: 10.1103/PhysRevA.55.900.

[39] Christopher A. Fuchs and Jeroen van de Graaf. Cryptographic distinguishability measures for quantum-mechanical states. *IEEE Transactions on Information Theory*, 45(4):1216–1227, 1999. doi: 10.1109/18.761271.

[40] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991.

[41] Eric Dennis, Alexei Kitaev, Andrew Landahl, and John Preskill. Topological quantum memory. *Journal of Mathematical Physics*, 43(9):4452–4505, 2002. doi: 10.1063/1.1499754.

[42] Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86:032324, 2012. doi: 10.1103/PhysRevA.86.032324.

[43] John Watrous. *The Theory of Quantum Information*. Cambridge University Press, Cambridge, 2018. doi: 10.1017/9781316848142.

[44] Howard Barnum, Claude Crépeau, Daniel Gottesman, Adam Smith, and Alain Tapp. Authentication of quantum messages. *arXiv preprint quant-ph/0205128*, 2002.

[45] Christopher Portmann. Quantum authentication with key recycling. In *EUROCRYPT 2017, Part III*, pages 339–368, 2017.